TAKE-HOME PROJECT

Advanced topics in Macroeconometric Forecasting with DSGE models_SS2024

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Question 1:

<u>*Summarize Data</u>

```
summarize if $insample
```

Variable	Obs	Mean	Std. dev.	Min	Max
ly	201	9.358302	.4044081	8.615784	9.975021
lc	201	8.925399	.4342952	8.132691	9.605677
pi	201	3.297647	2.284967	.1153194	10.37236
r	201	4.956119	4.0121	.07	19.1
time	201	146	58.16786	46	246

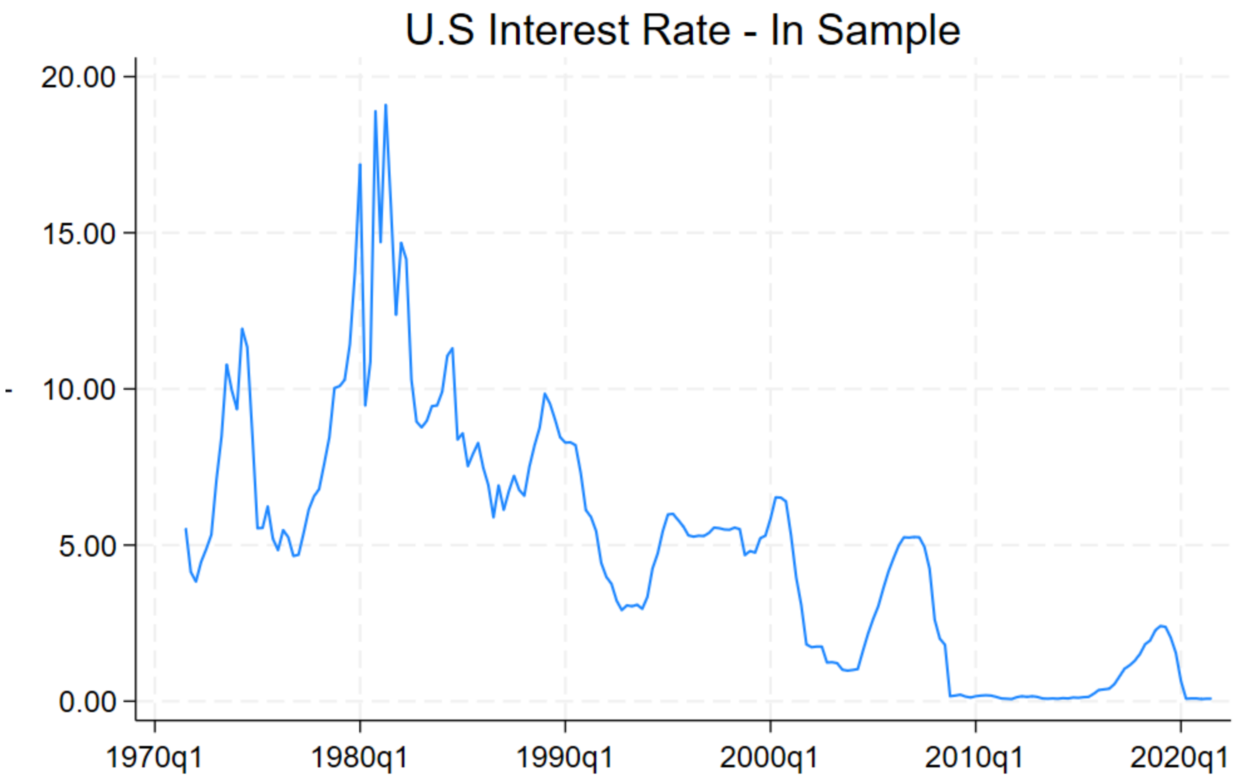
```
. summarize if $outsample
```

Variable	Obs	Mean	Std. dev.	Min	Max
ly	10	10.00498	.017315	9.985443	10.03231
lc	10	9.633712	.0157039	9.615487	9.659109
pi	10	5.006173	1.929073	2.397208	7.375045
r	10	3.387	2.18103	.08	5.33
time	10	251.5	3.02765	247	256

```
. summarize
```

Variable	Obs	Mean	Std. dev.	Min	Max
ly	217	9.366785	.4327216	8.5754	10.03231
lc	217	8.935277	.4654158	8.086058	9.659109
pi	217	3.426062	2.280142	.1153194	10.37236
r	217	4.908894	3.911392	.07	19.1
time	2,436	1257.5	703.357	40	2475

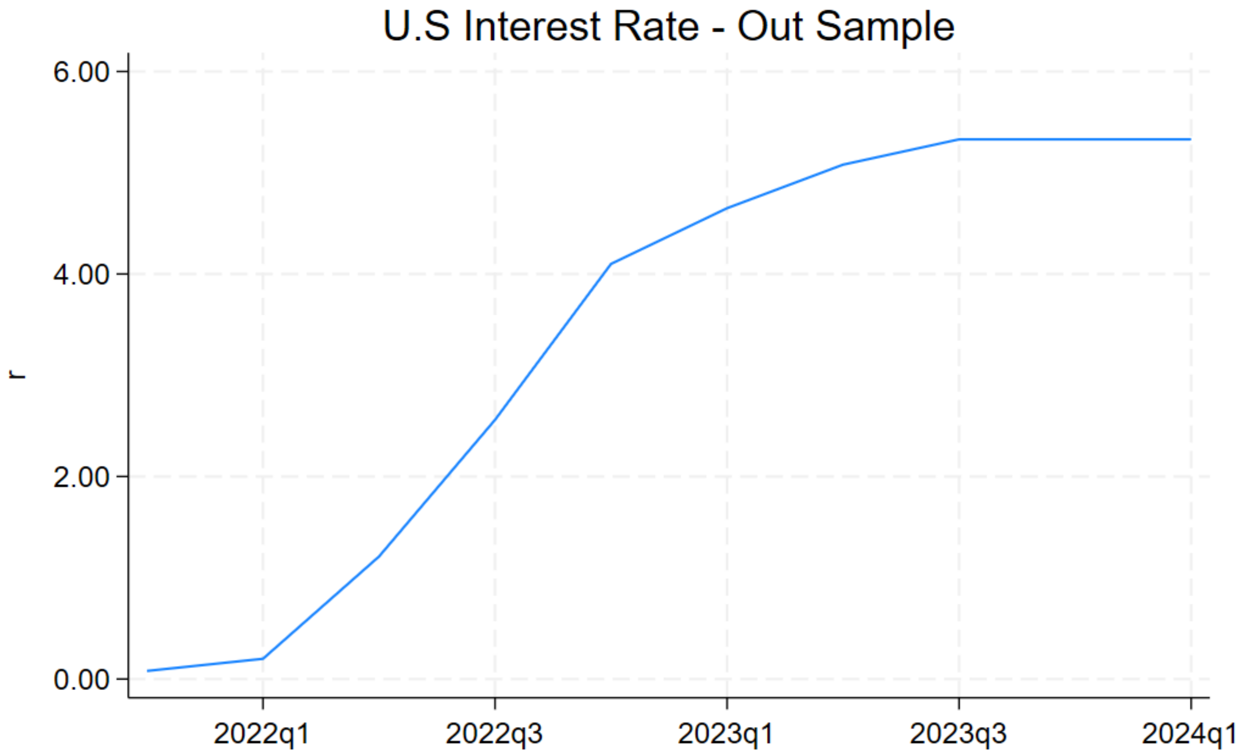
Graph U.S interest Rate for In-Sample data:



Comment:

- The United States experienced the high interest rates during the 1970s and 1980s, largely due to the impact of oil price shocks. Since then, interest rates have generally trended downwards, with some fluctuations, until reaching historic lows in the years of financial crisis.

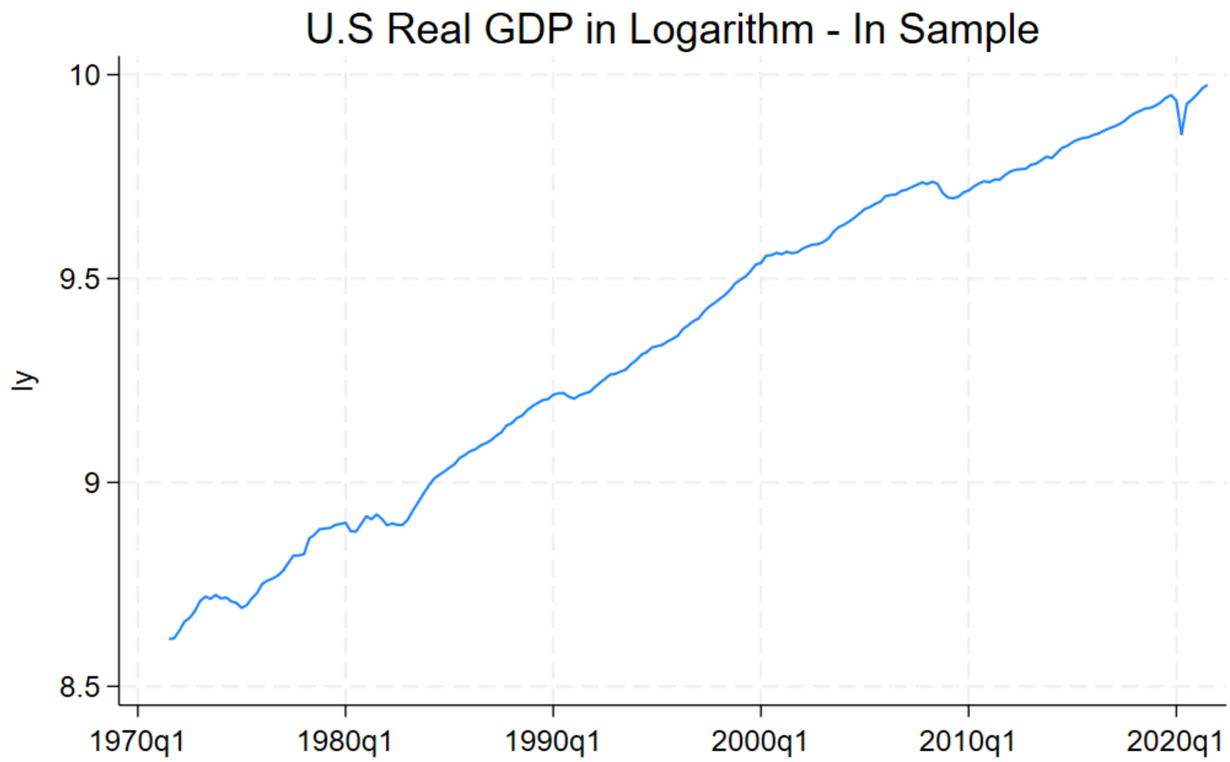
***U.S interest Rate for Out-Sample data:**



Comments:

Analyzing the U.S interest rate beyond the in-sample period (out-of-sample), we observe a tendency for it to revert back towards the in-sample average of around 4.95. This suggests the out-of-sample behavior aligns with the long-term average interest rate of 4.9 observed in the entire dataset.

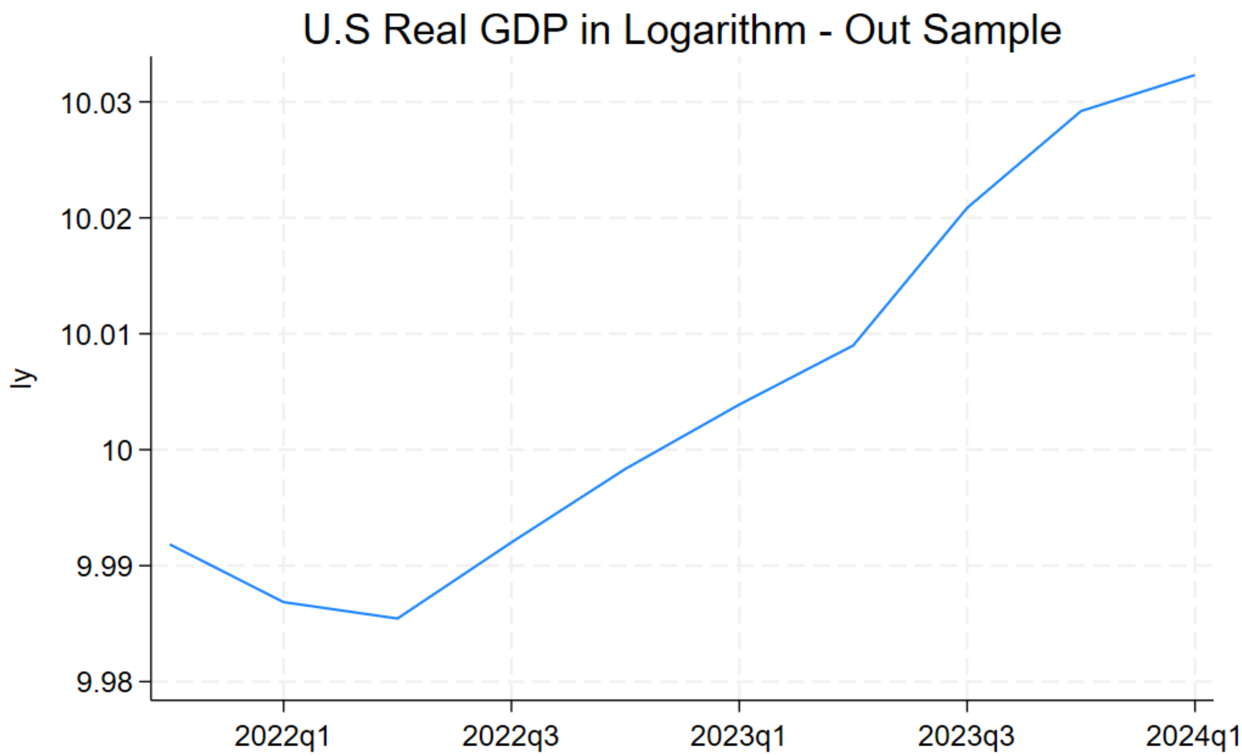
***Graph U.S Real GDP for In-Sample data**



Comments:

The real GDP data displays a **linear trend**, indicating non-stationarity. Interestingly, within the in-sample period, the data also exhibits a pattern reminiscent of a **random walk with drift**.

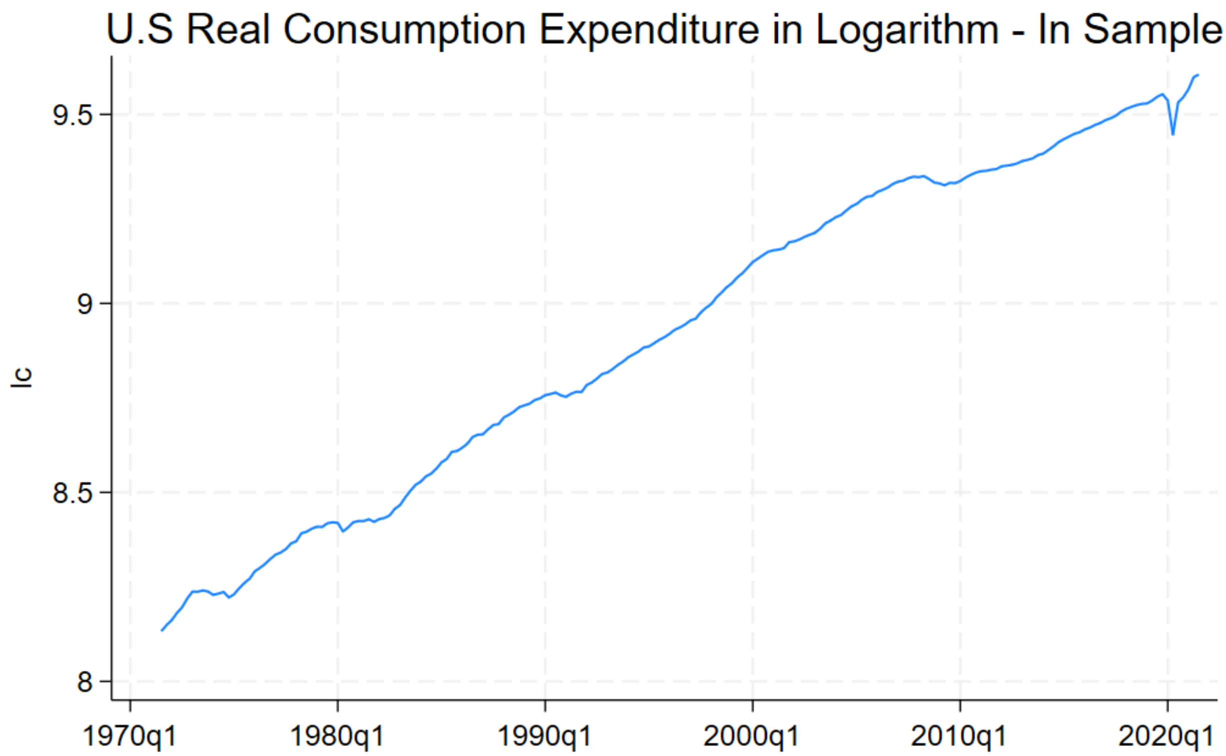
***Graph U.S Real GDP for Out-Sample data**



Comment:

The positive trend in real GDP observed during the in-sample period seems to be reflected in the out-of-sample period, suggesting sustained economic growth.

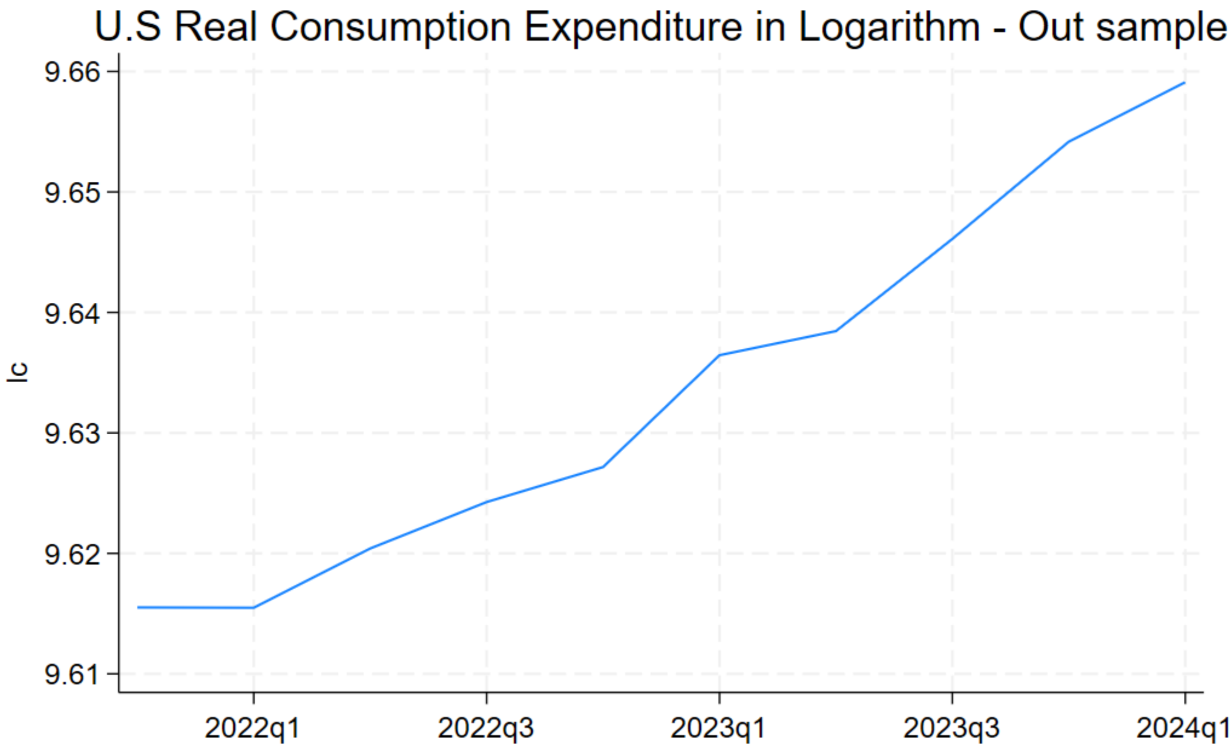
***Graph U.S Real Consumption Expenditure for In-Sample data**



Comment:

The graph reveals a time trend in consumption expenditure, indicating non-stationarity. However, in 2020, likely due to the COVID-19 pandemic, consumption sharply declined before reverting to the upward trend.

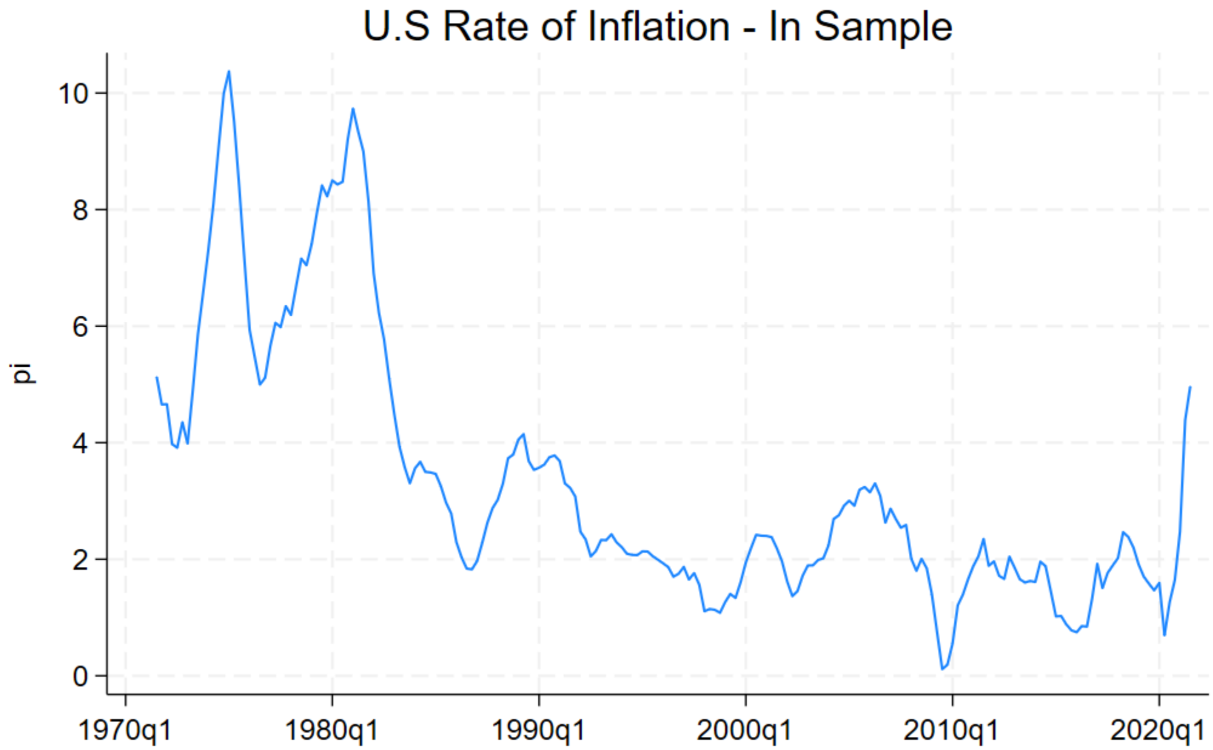
***Graph U.S Real Consumption Expenditure for Out-Sample data**



Comments:

The sharp increase in real consumption observed after the COVID-19 pandemic in the out-of-sample period (until 2024) is mirrored by the pattern within the in-sample period.

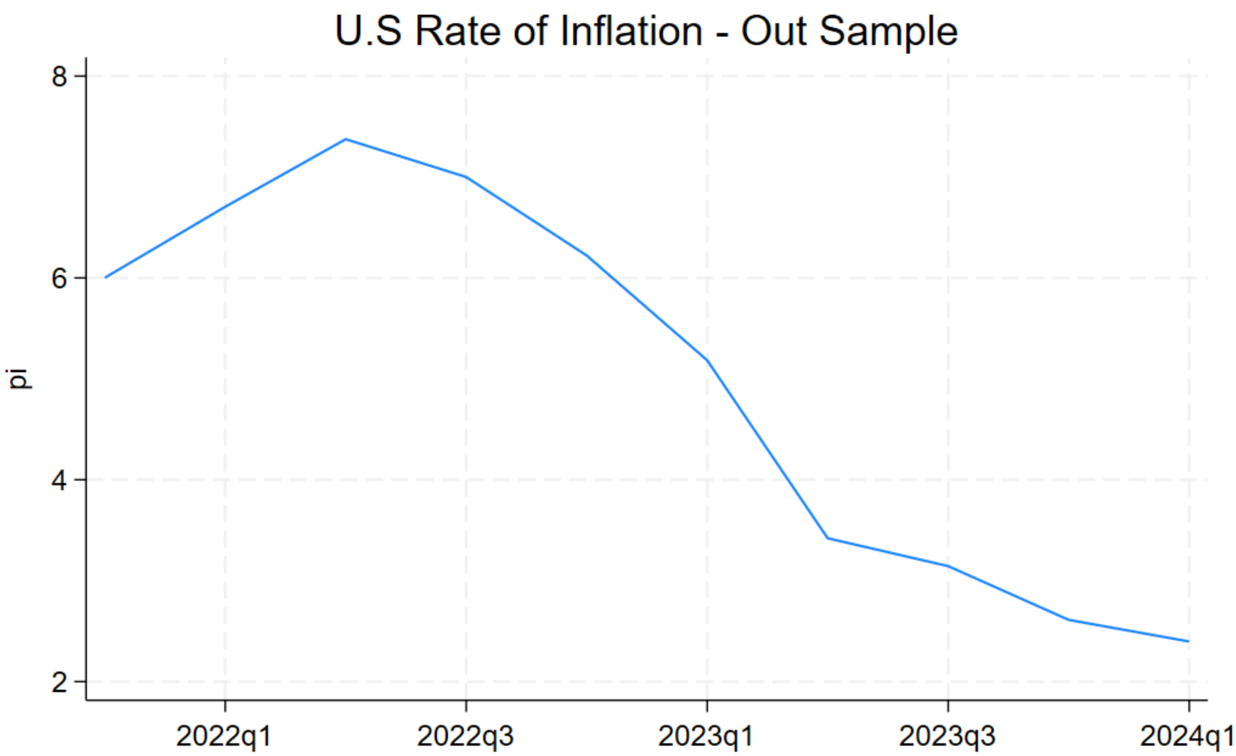
***U.S Inflation Rate _In Sample:**



Comment:

The 1970s and 1980s witnessed a surge in inflation due to oil price shocks. In COVID-19 pandemic, the inflation rose within the in-sample period, although it appears to have reverted back to the mean in the out-of-sample data.

***U.S Inflation Rate _Out Sample**



Comment:

- **Note for forecasting**: Based on the graph analysis, the in-sample data captures the underlying patterns observed in the out-of-sample data for each series (consumption, GDP, Inflation, interest rate). This suggests that we can choose models appropriate for these patterns, using the estimated coefficients from the in-sample data, to generate forecasts for the out-of-sample period or long-run period ahead.
- Forecasting GDP necessitates incorporating a comprehensive set of variables, including inflation, consumption, interest rates, and past GDP values. This is because these factors can have complex, interrelated effects. For example, rising consumption can stimulate GDP growth, while higher interest rates might dampen economic activity and lead to a decline in GDP.

Question 2:

```
. arimasoc ly if $insample, maxar(2) maxma(2)
Fitting models (9): .......xx done

Lag-order selection criteria

Sample: . thru .                                    Number of obs = 168
```

Model	LL	df	AIC	BIC	HQIC
ARMA(0,0)	-102.7339	2	209.4678	216.0744	212.1411
ARMA(0,1)	31.20552	3	-56.41105	-46.50113	-52.40107
ARMA(0,2)	149.6658	3	-293.3315	-283.4216	-289.3215
ARMA(1,0)	583.7324	3	-1161.465	-1151.555	-1157.455
ARMA(1,1)	589.2402	4	-1170.48	-1157.267	-1165.134
ARMA(1,2)	593.5895	5	-1177.179	-1160.662	-1170.496
ARMA(2,0)	591.8887	4	-1175.777	-1162.564	-1170.431
ARMA(2,1)					
ARMA(2,2)					

```
Note: Rows with missing values indicate models that failed to converge.
Selected (max) LL:   ARMA(1,2)
Selected (min) AIC:  ARMA(1,2)
Selected (min) BIC:  ARMA(2,0)
Selected (min) HQIC: ARMA(1,2)

.
end of do-file

.
```

Comment:

We can employ a Stata code like ***arimasoc*** to identify the optimal ARMA model order. This involves setting the maximum lags for the autoregressive (AR) and moving average (MA) components and choosing the ARMA (1,2) model based on the Akaike Information Criterion (AIC).

Subsequently, we can perform a regression analysis using the selected ARMA (1,2) model.

```
ARIMA regression

Sample: 1971q3 thru 2021q3                      Number of obs     =        201
                                                Wald chi2(3)      =  177023.52
Log likelihood = 593.5895                       Prob > chi2       =     0.0000
```

ly	Coefficient	OPG std. err.	z	P>|z|	[95% conf.	interval]
ly						
_cons	9.299263	.6769268	13.74	0.000	7.972511	10.62602
ARMA						
ar						
L1.	.9996821	.0041615	240.22	0.000	.9915258	1.007839
ma						
L1.	.1724683	.047496	3.63	0.000	.0793779	.2655587
L2.	.1917737	.095461	2.01	0.045	.0046736	.3788739
/sigma	.0123733	.0003314	37.33	0.000	.0117237	.0130229

```
Note: The test of the variance against zero is one sided, and the two-sided
      confidence interval is truncated at zero.
```

Interprete: The coefficient of the first lag of AR(1) in this model is 0,999 < 1, so it will be weakly stationary, we can use this to forecast the sample with the mean reverting pattern!

- **Model:** ARMA (1,2) - The model uses the previous value (lag 1) and two past error terms (moving average terms) to predict the current value.
- **Coefficient:**
- A coefficient of 0.9996 for the first lag (AR (1) term) suggests weak stationarity (statistically significant at 5%). Because it's less than 1 in absolute value, the impact of past values on the current value diminishes over time. So, the first lag of the GDP in this regression table have a positive impact on current GDP.

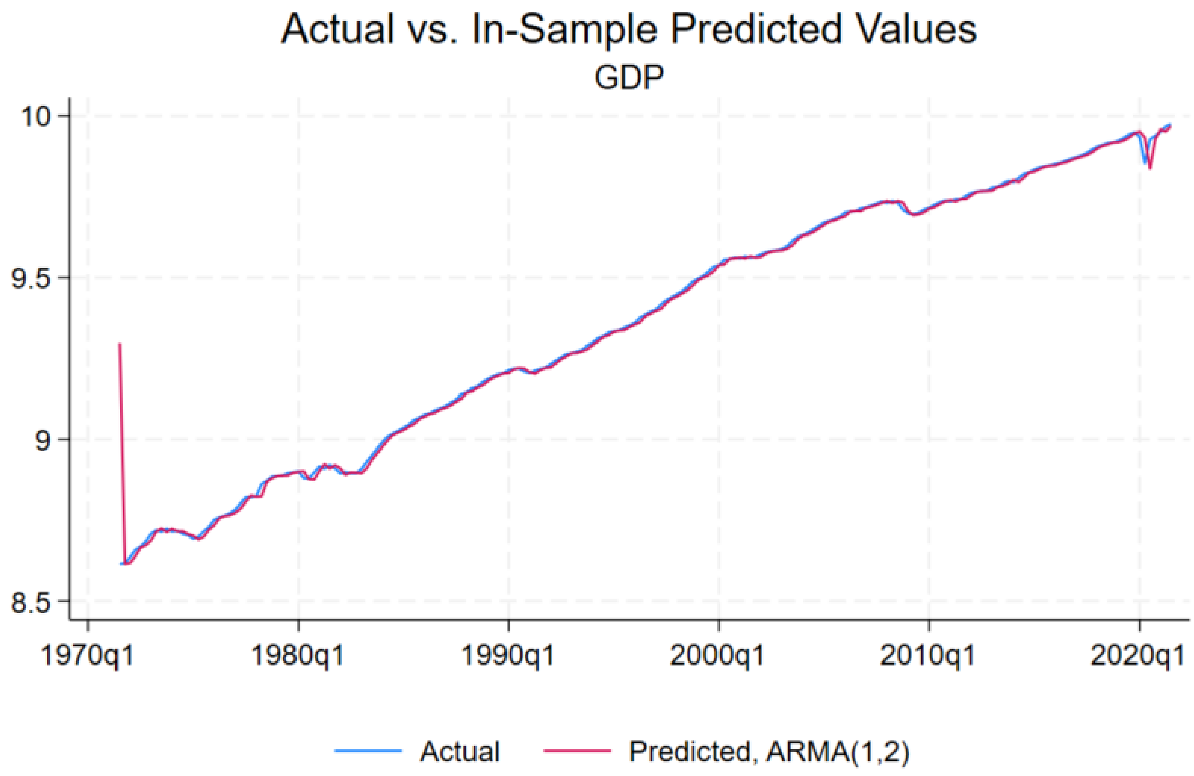
- The coefficient of the MA parts (0.1724, 0.1917, respectively) are all statistically significant at 5%.

- **Forecasting:** This weak stationarity allows the model to capture the mean reverting pattern in the data. The variable tends to return toward its average value after temporary fluctuations, making the model suitable for forecasting future values that will likely stay close to the mean.

Comment:

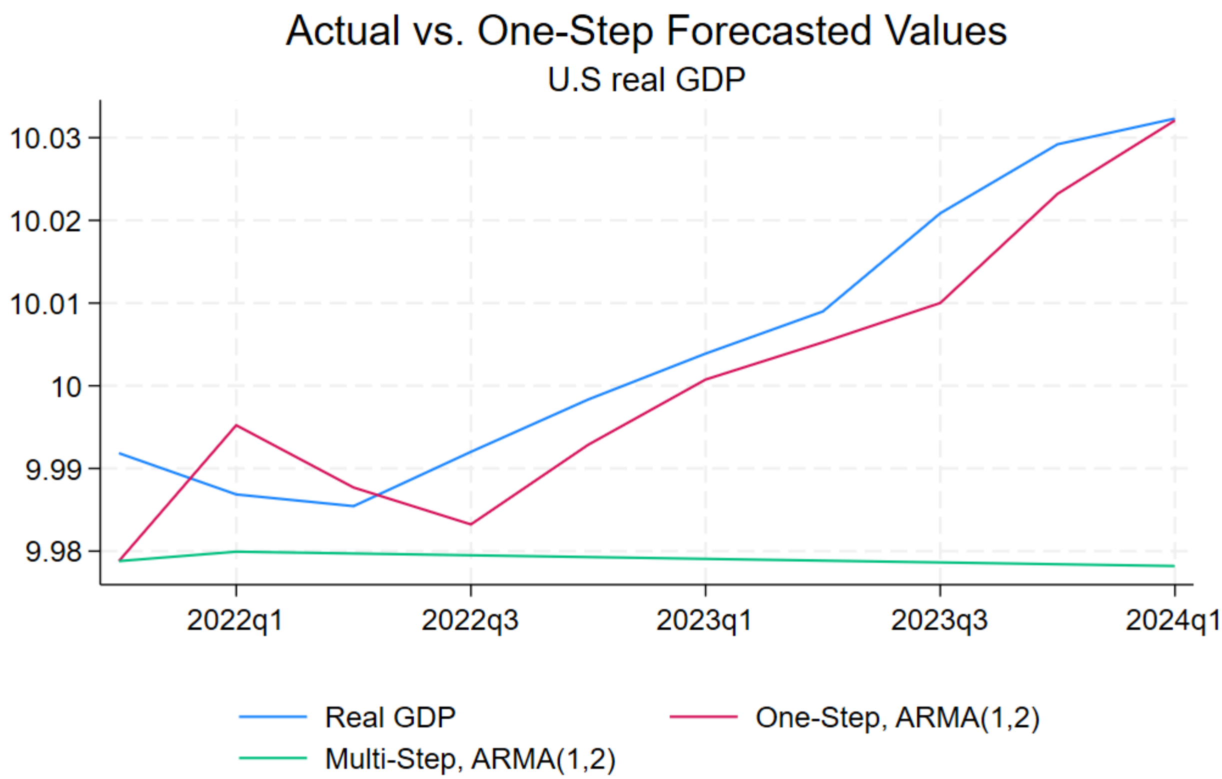
The coefficient for the first lag in this ARMA(1,2) model is 0.9996. This value, being less than 1 in absolute value, indicates weak stationarity. This characteristic allows the model to capture the mean reverting pattern in the data, making it suitable for forecasting future values that will likely return toward the average over time.

***In Sample- Fit:**



- The ARMA (1,2) model exhibits a good fit to the real data, as evidenced by the close correspondence between the model's predictions and the actual observations.

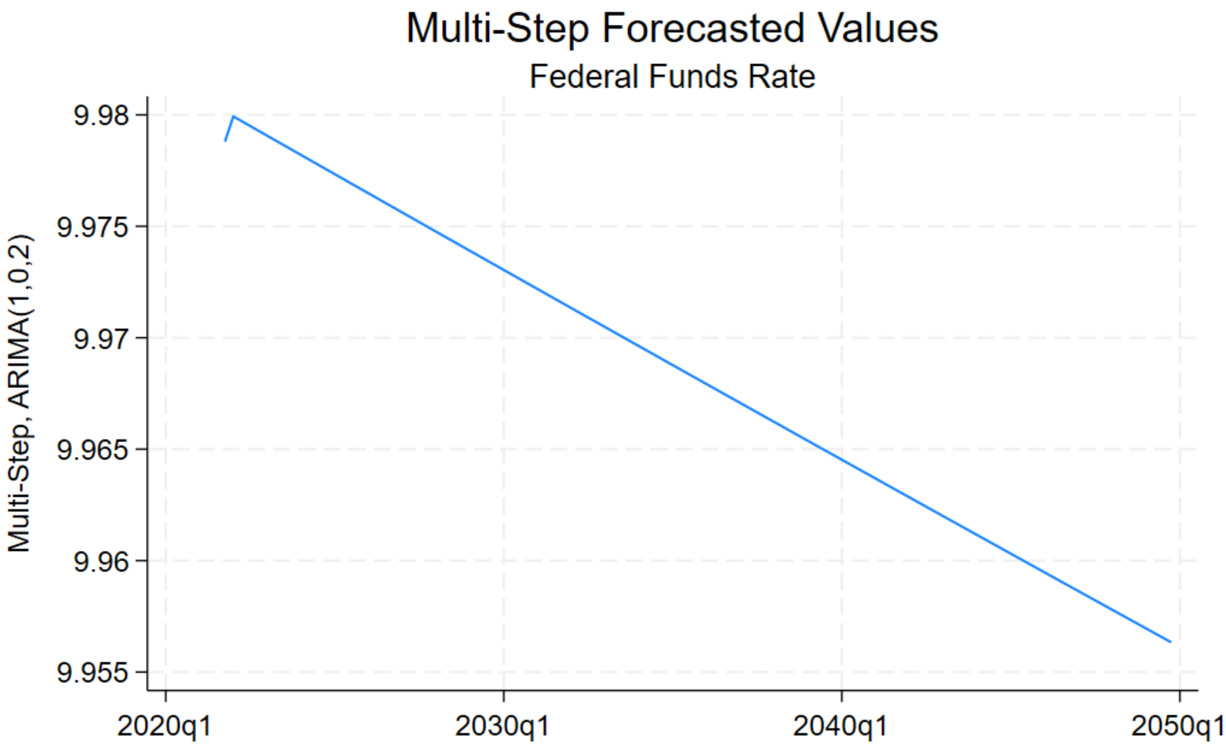
***Using the model ARMA (1,2) to forecast the Out-Sample Period and In Sample Period:**



Comment:

Given the mean-reverting properties of ARMA models, the multi-step forecast might converge towards the estimated mean value of the data.

Forecast:



Comment:

The multi-step ARMA forecast appears to have mean-reverting property even in the long term, with the model suggesting a return toward the mean value after 30 years long run ahead forcast.

***Alternatively, when I choose ARMA (3,0):**

```
. arimasoc ly if $insample, maxar(3) maxma(3)
Fitting models (16): .........xxx..xx done

Lag-order selection criteria

Sample: . thru .                               Number of obs = 167
```

Model	LL	df	AIC	BIC	HQIC
ARMA(0,0)	-102.7339	2	209.4678	216.0744	212.1411
ARMA(0,1)	31.20552	3	-56.41105	-46.50113	-52.40107
ARMA(0,2)	149.6658	3	-293.3315	-283.4216	-289.3215
ARMA(0,3)	240.9804	4	-473.9608	-460.7476	-468.6141
ARMA(1,0)	583.7324	3	-1161.465	-1151.555	-1157.455
ARMA(1,1)	589.2402	4	-1170.48	-1157.267	-1165.134
ARMA(1,2)	593.5895	5	-1177.179	-1160.662	-1170.496
ARMA(1,3)	596.717	6	-1181.434	-1161.614	-1173.414
ARMA(2,0)	591.8887	4	-1175.777	-1162.564	-1170.431
ARMA(2,1)					
ARMA(2,2)					
ARMA(2,3)					
ARMA(3,0)	598.2717	5	-1186.543	-1170.027	-1179.86
ARMA(3,1)	598.2465	6	-1184.493	-1164.673	-1176.473
ARMA(3,2)					
ARMA(3,3)					

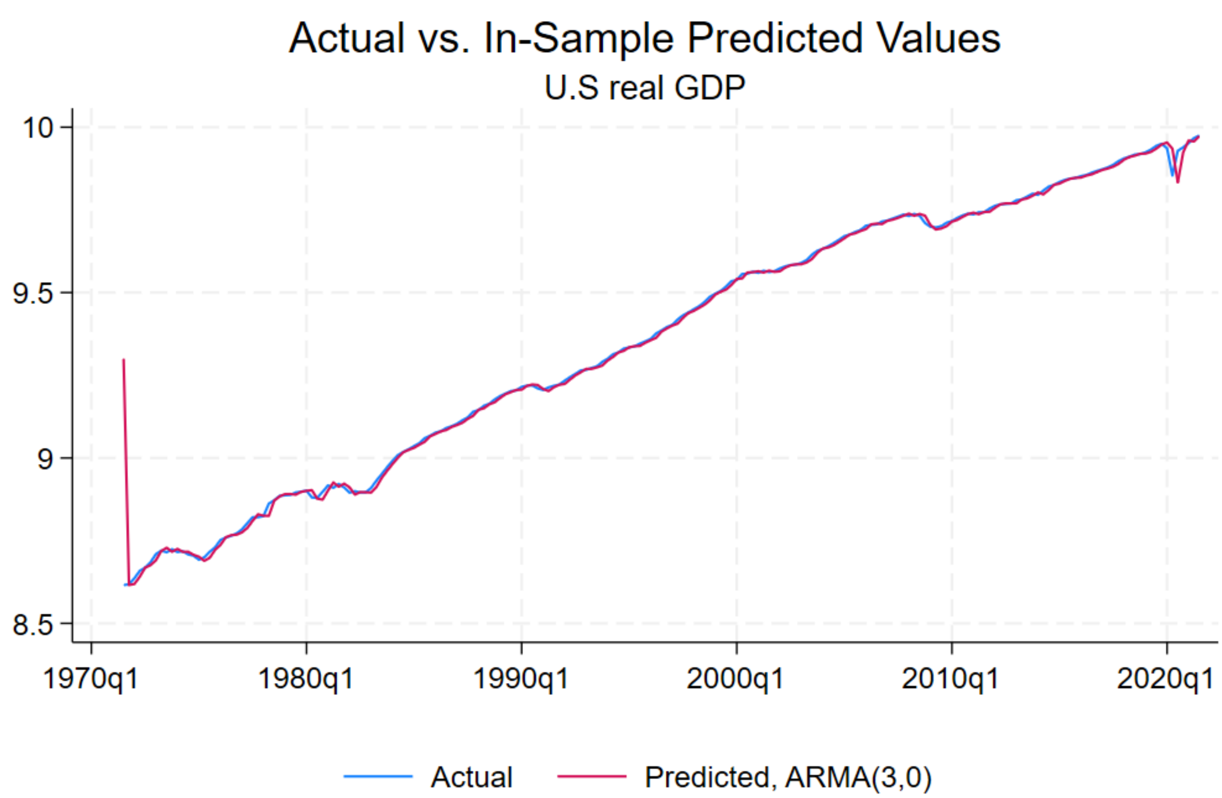
```
Note: Rows with missing values indicate models that failed to converge.
Selected (max) LL:   ARMA(3,0)
Selected (min) AIC:  ARMA(3,0)
Selected (min) BIC:  ARMA(3,0)
Selected (min) HQIC: ARMA(3,0)
```

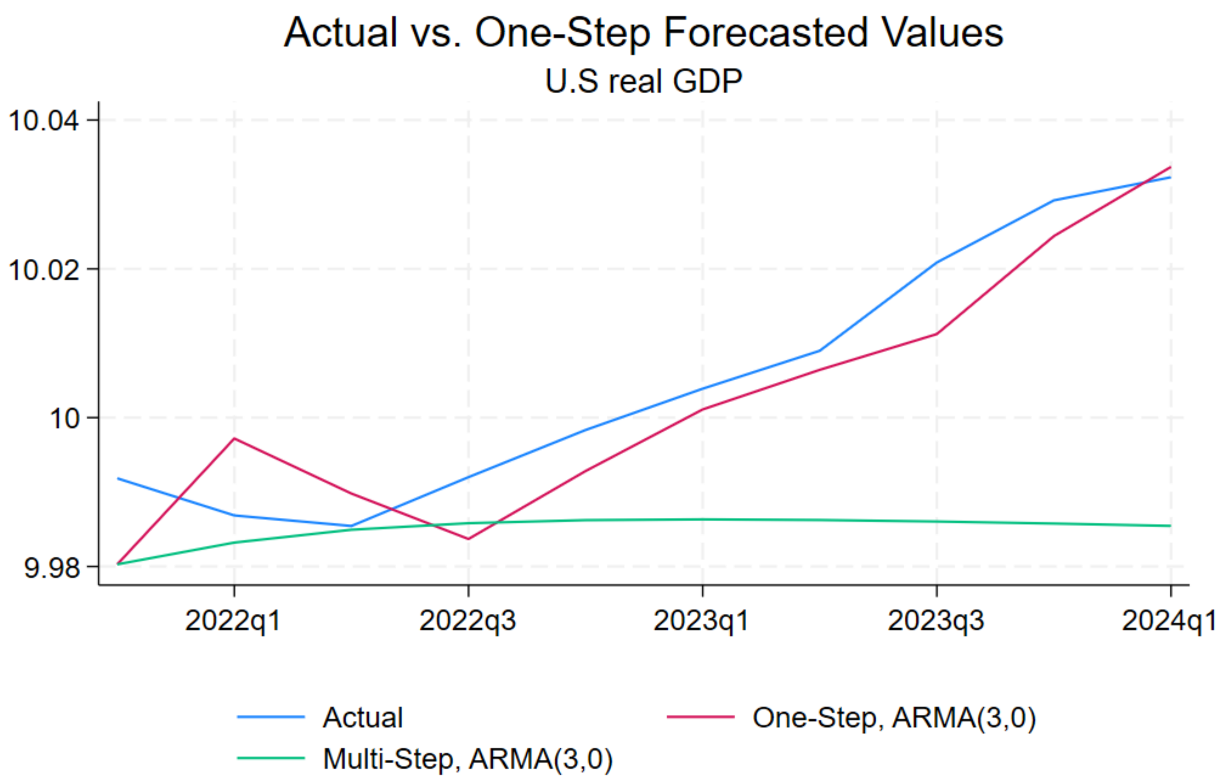
With the regression table for ARMA(3,0):

```
ARIMA regression

Sample: 1971q3 thru 2021q3                      Number of obs     =        201
                                                Wald chi2(3)      =  371637.80
Log likelihood = 598.2717                       Prob > chi2       =     0.0000

------------------------------------------------------------------------------
             |                 OPG
          ly | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
ly           |
       _cons |   9.300395   .6601875    14.09   0.000     8.006451    10.59434
-------------+----------------------------------------------------------------
ARMA         |
          ar |
         L1. |   1.210031   .0484356    24.98   0.000     1.115099    1.304963
         L2. |   .0391954   .0882936     0.44   0.657    -.1338569    .2122478
         L3. |  -.2495292   .0836003    -2.98   0.003    -.4133828   -.0856755
-------------+----------------------------------------------------------------
      /sigma |   .0120837   .0003383    35.72   0.000     .0114206    .0127467
------------------------------------------------------------------------------
Note: The test of the variance against zero is one sided, and the two-sided
      confidence interval is truncated at zero.
```





Comment:

- With ARMA(3,0) the forecast results are quite similar to the ARMA (1,2), so I choose ARMA(1,2) to compare with other forecasting models.

Question 3:

Lag-order selection criteria

Sample: **1971q3** thru **2021q3** Number of obs = **201**

Model	LL	df	AIC	BIC	HQIC
ARFIMA(0,0)	**268.9954**	**3**	**-531.9908**	**-522.0808**	**-527.9808**
ARFIMA(0,1)	**375.3237**	**4**	**-742.6475**	**-729.4343**	**-737.3009**
ARFIMA(0,2)	**442.2367**	**5**	**-874.4735**	**-857.957**	**-867.7902**
ARFIMA(0,3)	**479.1437**	**6**	**-946.2873**	**-926.4675**	**-938.2674**
ARFIMA(1,0)					
ARFIMA(1,1)					
ARFIMA(1,2)					
ARFIMA(1,3)					
ARFIMA(2,0)					
ARFIMA(2,1)					
ARFIMA(2,2)					
ARFIMA(2,3)					
ARFIMA(3,0)					
ARFIMA(3,1)					
ARFIMA(3,2)					
ARFIMA(3,3)					

Note: Rows with missing values indicate models that failed to converge.
Selected (max) LL: **ARFIMA(0,3)**
Selected (min) AIC: **ARFIMA(0,3)**
Selected (min) BIC: **ARFIMA(0,3)**
Selected (min) HQIC: **ARFIMA(0,3)**

Comment:

By leveraging the Akaike Information Criterion (AIC), we selected the ARFIMA(0,d,3) with d(-0.5, 0.5) model for both estimation and forecasting. This choice of model captures the presence of long memory property in the data, which is a characteristic not captured by ARIMA models.

Estimation Regression Table:

```
ARFIMA regression

Sample: 1970q1 thru 2024q1                      Number of obs =        217
                                                Wald chi2(4)  = 92971.10
Log likelihood = 513.11837                      Prob > chi2   =     0.0000
```

ly	Coefficient	Std. err.	z	P>|z|	[95% conf. interval]	
ly						
_cons	9.348133	1.010803	9.25	0.000	7.366995	11.32927
ARFIMA						
ma						
L1.	1.263594	.0695842	18.16	0.000	1.127212	1.399977
L2.	1.096679	.0664732	16.50	0.000	.966394	1.226964
L3.	.5642082	.0475169	11.87	0.000	.4710768	.6573396
d	.4988284	.0016513	302.07	0.000	.4955918	.5020649
/sigma2	.0004936	.0000475	10.39	0.000	.0004005	.0005867

```
Note: The test of the variance against zero is one sided, and the two-sided
      confidence interval is truncated at zero.
```

Comment:

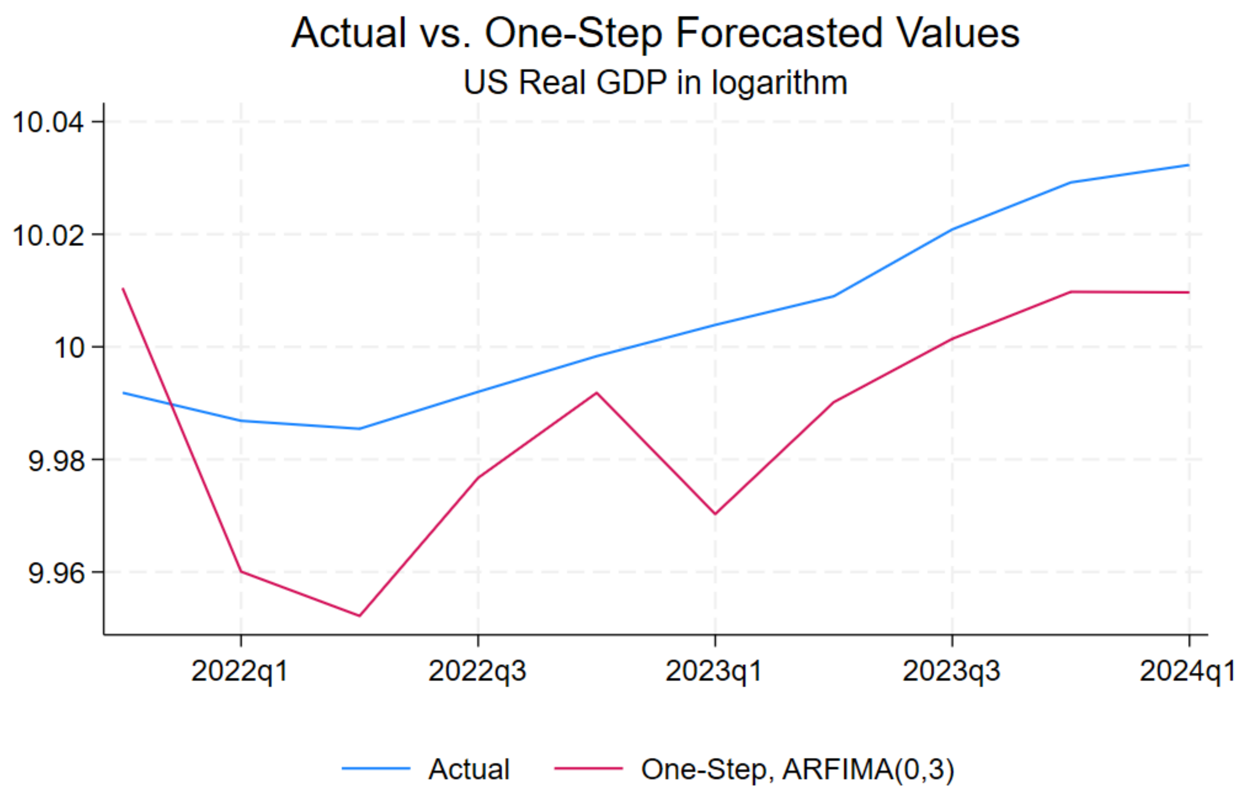
The unique aspect of ARFIMA is the fractional differencing parameter (d), which allows for non-integer differencing. In traditional ARIMA models, differencing is done with whole numbers (integers) to make the series stationary. In ARFIMA, the differencing parameter can take fractional values, allowing for a more flexible treatment of long-range dependence or memory in the time series.

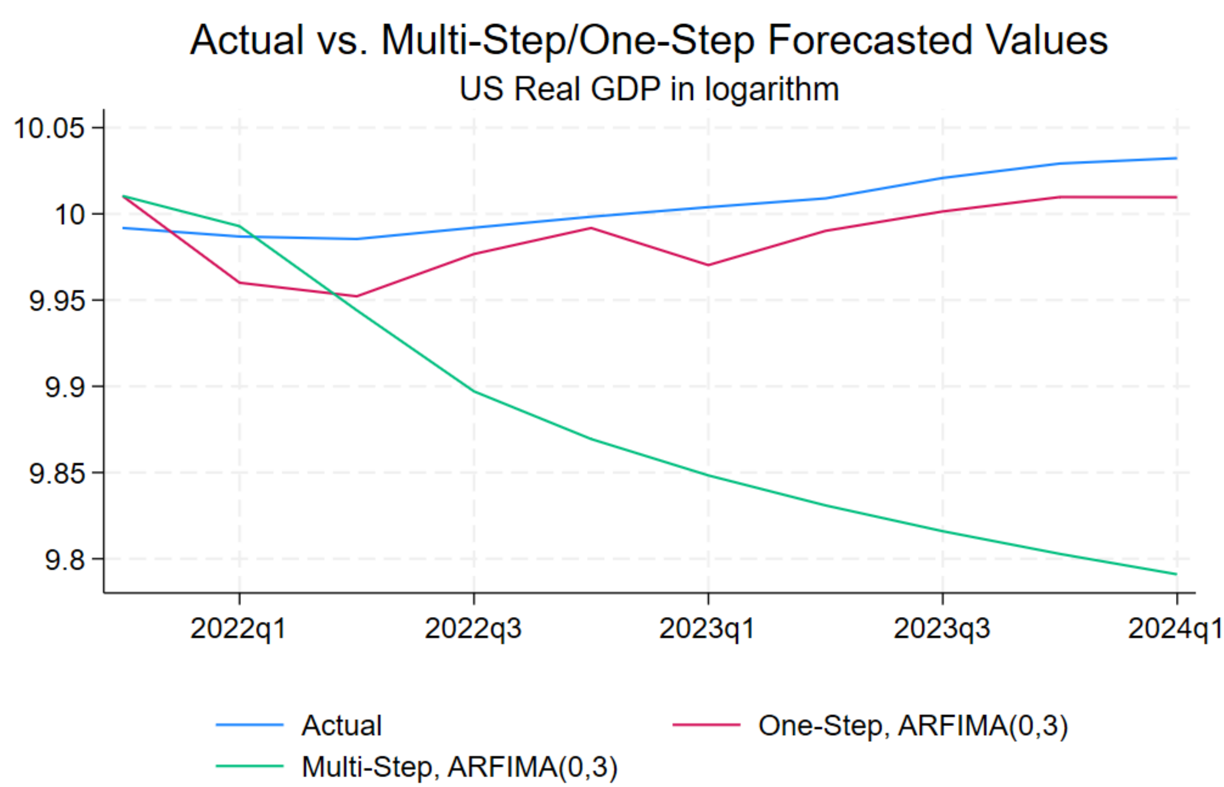
ARFIMA models can exhibit mean-reverting behavior similar to ARMA (Autoregressive Moving Average) models, especially when the differencing parameter 'd' is less than 0.5.

In ARFIMA models, the fractional differencing parameter 'd' determines the degree of mean reversion. When 'd' is less than 0.5, it implies that the series is mean reverting, meaning it tends to return toward its mean after experiencing shocks or deviations.

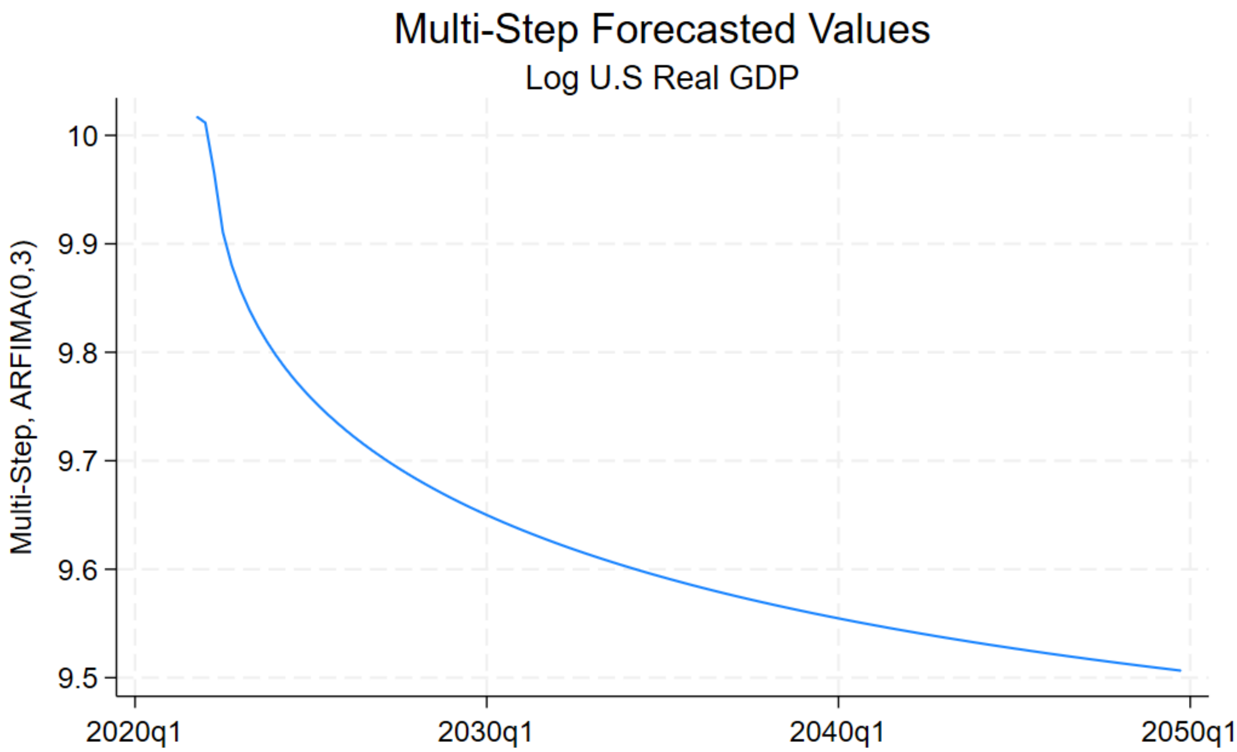
Coefficient: A coefficient of MA parts (1.263359, 1.0966679, 0.56420) are all statistically significant at 5% representing how much the model dampens shocks to the series.

*** Forecasting for ARFIMA(0,0,3) Model:**


Actual vs. One-Step Forecasted Values
US Real GDP in logarithm
10.04
10.02
10
9.98
9.96
2022q1
2022q3
2023q1
2023q3
2024q1
Actual
One-Step, ARFIMA(0,3)


Actual vs. Multi-Step/One-Step Forecasted Values
US Real GDP in logarithm
10.05
10
9.95
9.9
9.85
9.8
2022q1
2022q3
2023q1
2023q3
2024q1
Actual
One-Step, ARFIMA(0,3)
Multi-Step, ARFIMA(0,3)

* Forecasting for Extended Out-of-Sample Period:



Comments:

This means that even 30 years later, the ARFIMA model can still demonstrate a tendency to return to its long-term average or mean-value after experiencing shocks or deviations.

This is because lower values of 'd' indicate a stronger mean-reverting behavior, akin to a higher degree of stationarity in the series.

Question 4:

```
. arimasoc d.ly if $insample, maxar(3) maxma(3)
Fitting models (16): ................ done

Lag-order selection criteria

Sample: 1971q3 thru 2021q3                    Number of obs = 201
```

Model	LL	df	AIC	BIC	HQIC
ARMA(0,0)	620.2423	2	-1236.485	-1229.878	-1233.811
ARMA(0,1)	620.2517	3	-1234.503	-1224.594	-1230.493
ARMA(0,2)	620.5298	4	-1233.06	-1219.846	-1227.713
ARMA(0,3)	620.6946	5	-1231.389	-1214.873	-1224.706
ARMA(1,0)	620.2527	3	-1234.505	-1224.595	-1230.495
ARMA(1,1)	620.2476	4	-1232.495	-1219.282	-1227.149
ARMA(1,2)	620.654	5	-1231.308	-1214.791	-1224.625
ARMA(1,3)	620.6973	6	-1229.395	-1209.575	-1221.375
ARMA(2,0)	620.5359	4	-1233.072	-1219.859	-1227.725
ARMA(2,1)	620.6338	5	-1231.268	-1214.751	-1224.584
ARMA(2,2)	622.8307	6	-1233.661	-1213.842	-1225.642
ARMA(2,3)	621.6139	7	-1229.228	-1206.105	-1219.871
ARMA(3,0)	620.6935	5	-1231.387	-1214.87	-1224.704
ARMA(3,1)	620.6936	6	-1229.387	-1209.567	-1221.367
ARMA(3,2)	622.214	7	-1230.428	-1207.305	-1221.071
ARMA(3,3)	622.0436	8	-1228.087	-1201.661	-1217.394

```
Selected (max) LL:   ARMA(2,2)
Selected (min) AIC:  ARMA(0,0)
Selected (min) BIC:  ARMA(0,0)
Selected (min) HQIC: ARMA(0,0)
```

Comment:

I employed **arimasoc** to determine the optimal model order for an ARIMA model, and then I ultimately chose the ARIMA (0,0) model for estimation and forecasting. This ARIMA(0,0) model implies a random walk process for the data.

```
. arima ly if $insample, arima(0,1,0)

(setting optimization to BHHH)
Iteration 0:   Log likelihood =  620.24231
Iteration 1:   Log likelihood =  620.24231

ARIMA regression

Sample: 1971q3 thru 2021q3                      Number of obs     =        201
                                                Wald chi2(.)      =          .
Log likelihood = 620.2423                       Prob > chi2       =          .
```

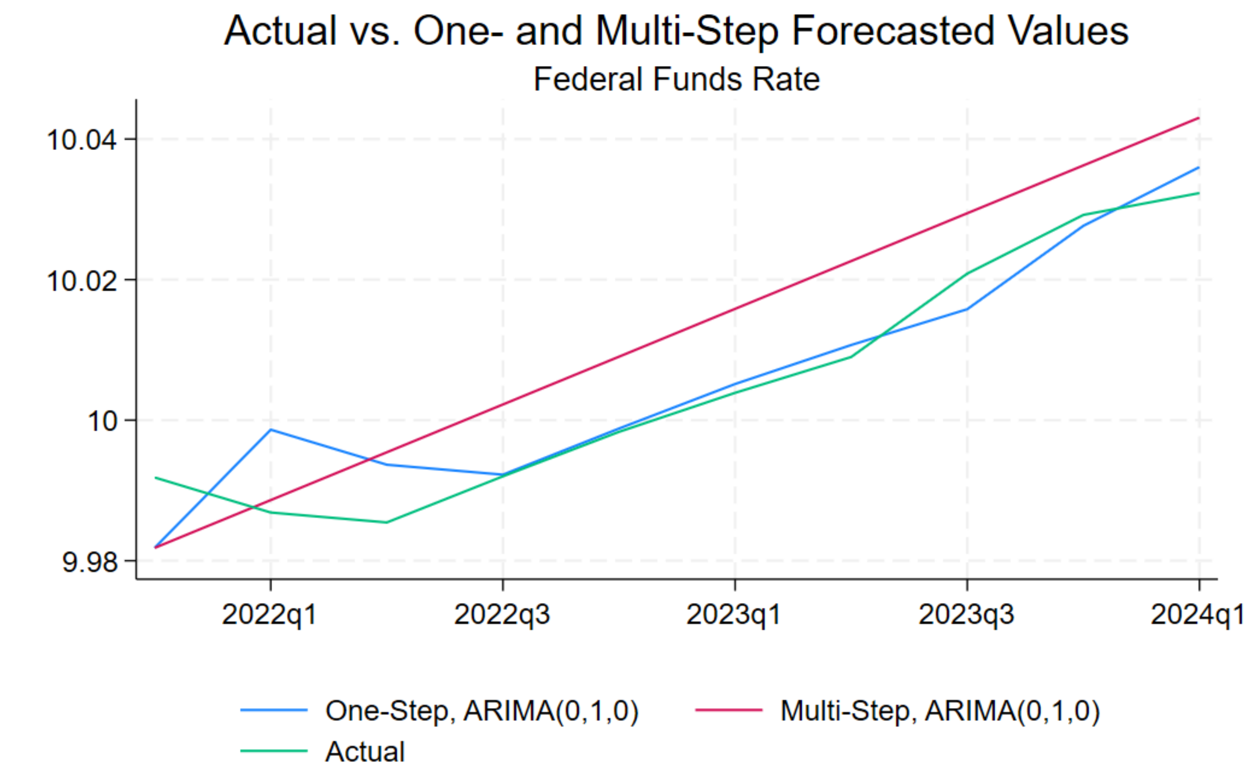
D.ly	Coefficient	OPG std. err.	z	P>|z|	[95% conf.	interval]
ly						
_cons	.0068031	.0008168	8.33	0.000	.0052022	.0084041
/sigma	.0110567	.0001535	72.02	0.000	.0107558	.0113576

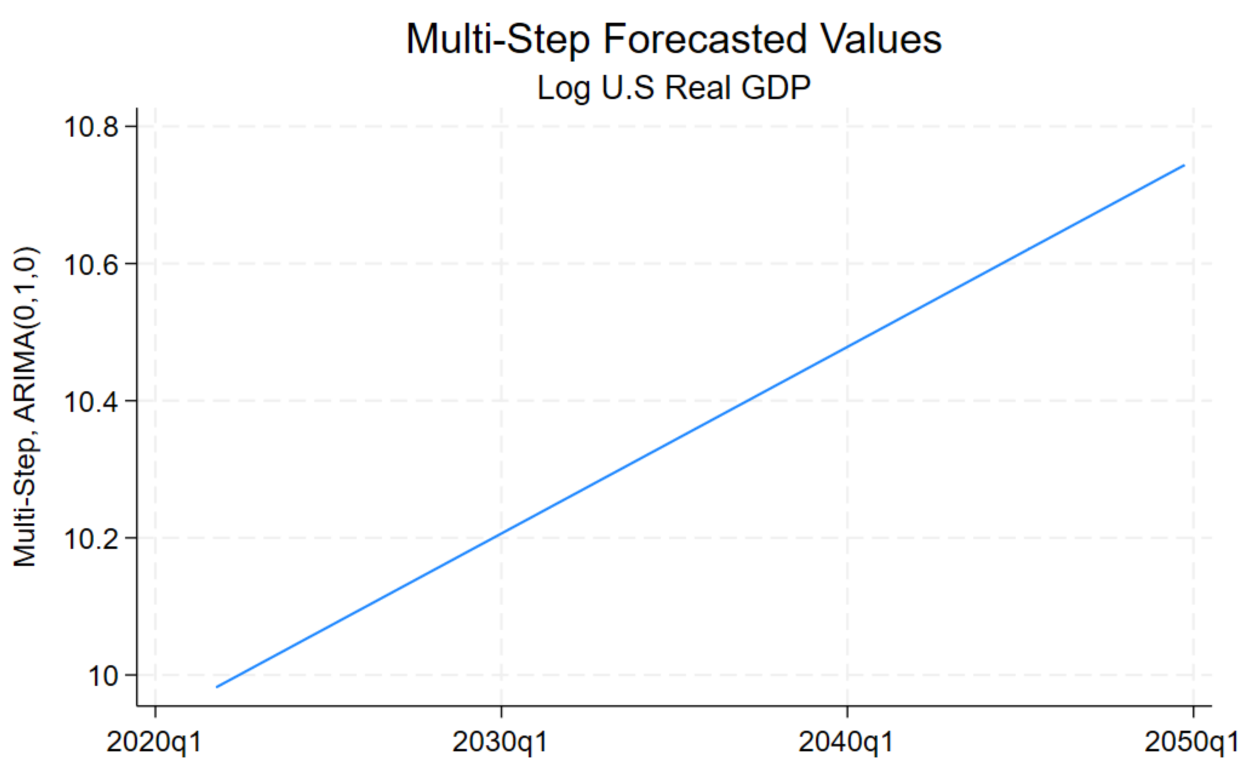
```
Note: The test of the variance against zero is one sided, and the two-sided
      confidence interval is truncated at zero.
```

Comment:

The regression analysis reveals a statistically significant positive constant term (coefficient) of 0.0068031 (statistically significant at 5% level).

This suggests a "random walk with drift" model, where the variable exhibits a positive upward trend over time. Additionally, the positive drift implies that the last observations in the in-sample period might have a permanent impact on the out-of-sample forecast, potentially causing it to exhibit a positive trend as well.


Actual vs. One- and Multi-Step Forecasted Values
Federal Funds Rate
10.04
10.02
10
9.98
2022q1
2022q3
2023q1
2023q3
2024q1
One-Step, ARIMA(0,1,0)
Multi-Step, ARIMA(0,1,0)
Actual


Multi-Step Forecasted Values
Log U.S Real GDP
Multi-Step, ARIMA(0,1,0)
10.8
10.6
10.4
10.2
10
2020q1
2030q1
2040q1
2050q1

The analysis reveals two distinct forecast patterns in the data. The ARIMA model, in contrast to AR, ARMA, and ARFIMA, does not exhibit mean reversion, where the forecast eventually returns towards the average value. This behavior in the out-of-sample data suggests a permanent upward shift. The positive coefficient of the intercept term in the regression table potentially contributes to this shift, indicating a permanent impact from the last observations (~9,975) of the in-sample period.

Question 6:

Variable R:

```
. * Lag Length Selection
. varsoc r if $insample, maxlag(6)

Lag-order selection criteria

   Sample: 1971q3 thru 2021q3                       Number of obs = 201
```

Lag	LL	LR	df	p	FPE	AIC	HQIC	SBIC
0	-563.958				16.177	5.62147	5.62812	5.6379
1	-328.692	470.53	1	0.000	1.57238	3.29046	3.30376	3.32333
2	-327.417	2.55	1	0.110	1.56808	3.28773	3.30768	3.33703
3	-324.159	6.5156	1	0.011	1.53325	3.26526	3.29186	3.331
4	-318.352	11.614	1	0.001	1.46165	3.21743	3.25068	3.29961
5	-317.674	1.3569	1	0.244	1.46635	3.22063	3.26053	3.31924
6	-309.721	15.906*	1	0.000	1.36834*	3.15145*	3.198*	3.26649*

```
   * optimal lag
   Endogenous: r
    Exogenous: _cons
```

I chose lag 6 using AIC information criteria!

```
. dfuller r if $insample, lags(5) regress

Augmented Dickey-Fuller test for unit root

Variable: r                              Number of obs  = 201
                                         Number of lags =   5

H0: Random walk without drift, d = 0
```

	Test statistic	Dickey-Fuller critical value 1%	5%	10%
Z(t)	-2.307	-3.476	-2.883	-2.573

MacKinnon approximate *p*-value for Z(t) = **0.1699**.

Regression table

D.r	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
r						
L1.	-.0497844	.021584	-2.31	0.022	-.0923537	-.007215
LD.	-.0382071	.0696868	-0.55	0.584	-.1756481	.099234
L2D.	-.1990472	.069062	-2.88	0.004	-.3352559	-.0628386
L3D.	.278957	.0685272	4.07	0.000	.143803	.414111
L4D.	-.0513262	.0685671	-0.75	0.455	-.1865588	.0839064
L5D.	.2733682	.0683932	4.00	0.000	.1384785	.4082578
_cons	.2355601	.1361355	1.73	0.085	-.0329355	.5040557

- The Augmented Dickey-Fuller (ADF) test failed to reject the null hypothesis (H0) at the 1% significance level (p_value= 0.1699). This suggests that the data series might exhibit a random walk without drift.

```
. dfgls r if $insample, maxlag(6) notrend

DF-GLS test for unit root                    Number of obs = 194
Variable: r
Lag selection: User specified                Maximum lag   =   6

                                  ———————— Critical value ————————
 [lags]        DF-GLS mu          1%            5%           10%
————————————————————————————————————————————————————————————————————
     6           -2.150        -2.587        -2.008       -1.696
     5           -2.264        -2.587        -2.013       -1.701
     4           -1.618        -2.587        -2.018       -1.706
     3           -1.821        -2.587        -2.024       -1.710
     2           -1.338        -2.587        -2.028       -1.715
     1           -1.712        -2.587        -2.033       -1.719
————————————————————————————————————————————————————————————————————
Opt lag (Ng-Perron seq t) = 5 with RMSE = 1.137119
Min SIC  =  .4199184 at lag 5 with RMSE = 1.137119
Min MAIC =  .3701245 at lag 5 with RMSE = 1.137119
```

- The Dickey-Fuller GLS test failed to reject the null hypothesis (H_0) of a unit root being present in the data at 1% significant level.

2. Variable Pi:

```
. varsoc pi if $insample, maxlag(6)
```

Lag-order selection criteria

Sample: **1971q3** thru **2021q3** Number of obs = **201**

Lag	LL	LR	df	p	FPE	AIC	HQIC	SBIC
0	-450.802				5.24705	4.49554	4.50219	4.51198
1	-105.43	690.75	1	0.000	.170516	1.06895	1.08225	1.10182
2	-65.6963	79.466	1	0.000	.115981	.683546	.703496	.732849
3	-64.004	3.3847	1	0.066	.115185	.676657	.703257	.742394
4	-63.9312	.14552	1	0.703	.116253	.685883	.719133	.768055
5	-53.3573	21.148	1	0.000	.105691	.59062	.63052	.689226*
6	-51.2836	4.1475*	1	0.042	.104569*	.579936*	.626486*	.694977

* optimal lag
Endogenous: **pi**
Exogenous: **_cons**

I chose 6 lags using AIC information criteria!

Variable: **pi** Number of obs = **201**
Number of lags = **5**

H0: Random walk without drift, d = 0

	Test statistic	Dickey-Fuller critical value 1%	5%	10%
Z(t)	-2.207	-3.476	-2.883	-2.573

MacKinnon approximate *p*-value for Z(t) = **0.2038**.

Regression table

D.pi	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
pi						
L1.	-.0229544	.0104023	-2.21	0.029	-.0434706	-.0024382
LD.	.5469926	.0706254	7.74	0.000	.4077004	.6862848
L2D.	.1342285	.0792611	1.69	0.092	-.0220957	.2905526
L3D.	.1598457	.0793006	2.02	0.045	.0034438	.3162477
L4D.	-.4016081	.0787983	-5.10	0.000	-.5570194	-.2461968
L5D.	.1504556	.0748115	2.01	0.046	.0029074	.2980039
_cons	.0768636	.041436	1.85	0.065	-.0048594	.1585865

The Augmented Dickey-Fuller (ADF) test statistic in the table is -2.207. At a significance level (1%, 5%, or 10%), we fail to reject the null hypothesis (H_0) of a unit root being present in the data (p_value = 0.2038). This suggests the series exhibits a random walk without drift.

```
. dfgls pi if $insample, maxlag(6) notrend

DF-GLS test for unit root                    Number of obs = 194
Variable: pi
Lag selection: User specified                Maximum lag   =   6

                                 ───────── Critical value ──────────
 [lags]       DF-GLS mu           1%            5%           10%
─────────────────────────────────────────────────────────────────────
      6          -1.709        -2.587        -2.008        -1.696
      5          -1.777        -2.587        -2.013        -1.701
      4          -1.559        -2.587        -2.018        -1.706
      3          -2.255        -2.587        -2.024        -1.710
      2          -2.257        -2.587        -2.028        -1.715
      1          -2.062        -2.587        -2.033        -1.719
─────────────────────────────────────────────────────────────────────
Opt lag (Ng-Perron seq t) = 5 with RMSE = .3078562
Min SIC  = -2.203179 at lag 4 with RMSE = .3105302
Min MAIC = -2.269718 at lag 4 with RMSE = .3105302

.
end of do-file
```

Comment:

The Dickey-Fuller GLS (DF-GLS) test, with the Modified AIC (MAIC) criterion, identified 4 lags as optimal for the model. However, the test failed to reject the null hypothesis (H_0) of a unit root at the 1% significance level. This suggests that the data series might still exhibit a unit root, even after considering lag structure.

***Variable ly:**

```
. varsoc ly if $insample, maxlag(6)

Lag-order selection criteria

   Sample: 1971q3 thru 2021q3                          Number of obs = 201
```

Lag	LL	LR	df	p	FPE	AIC	HQIC	SBIC
0	-102.734				.16436	1.03218	1.03883	1.04861
1	621.475	1448.4*	1	0.000	.000123*	-6.16393*	-6.15063*	-6.13106*
2	621.475	.00019	1	0.989	.000124	-6.15398	-6.13403	-6.10468
3	621.664	.37764	1	0.539	.000125	-6.14591	-6.11931	-6.08017
4	621.731	.13367	1	0.715	.000127	-6.13662	-6.10337	-6.05445
5	621.731	.00023	1	0.988	.000128	-6.12667	-6.08677	-6.02807
6	621.805	.14866	1	0.700	.000129	-6.11746	-6.07091	-6.00242

```
   * optimal lag
   Endogenous: ly
    Exogenous: _cons
```

Based on the Akaike Information Criterion (AIC), a VAR(1) model appears to be the most suitable choice for the data.

```
. dfuller ly if $insample, lags(0) regress

Dickey-Fuller test for unit root          Number of obs  = 201
Variable: ly                              Number of lags =   0

H0: Random walk without drift, d = 0
```

	Test statistic	Dickey-Fuller critical value 1%	5%	10%
Z(t)	-1.567	-3.476	-2.883	-2.573

```
MacKinnon approximate p-value for Z(t) = 0.5001.

Regression table
```

D.ly	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
ly L1.	-.0030181	.001926	-1.57	0.119	-.006816	.0007798
_cons	.0350271	.0180274	1.94	0.053	-.0005222	.0705764

The Dickey-Fuller test statistic is -1.567. At a significance level (1%, 5%, or 10%), we failed to reject the null hypothesis (H_0) of a unit root being present in the data.

```
. dfgls ly if $insample, maxlag(4) notrend

DF-GLS test for unit root                    Number of obs = 196
Variable: ly
Lag selection: User specified                Maximum lag   =   4

                                 ───────── Critical value ─────────
 [lags]        DF-GLS mu          1%             5%            10%
───────────────────────────────────────────────────────────────────
      4            2.727      -2.587         -2.018         -1.705
      3            3.038      -2.587         -2.023         -1.709
      2            3.599      -2.587         -2.028         -1.714
      1            4.532      -2.587         -2.032         -1.718
───────────────────────────────────────────────────────────────────
Opt lag (Ng-Perron seq t) = 3 with RMSE =   .01158
Min SIC   = -8.817173 at lag 2 with RMSE = .0116905
Min MAIC  = -8.769434 at lag 4 with RMSE = .0115395
```

Comment:

With DF-GLS, The Modified Akaike Information Criterion (MAIC) suggests using 4 lags for model estimation. However, the results from the unit root test led us to reject the null hypothesis (presence of a unit root) at the 1% significance level. This implies that the data series might be stationary after considering a lag structure of 4.

<u>*Variable lc:</u>

```
. varsoc lc if $insample, maxlag(6)
```

Lag-order selection criteria

Sample: 1971q3 thru 2021q3 Number of obs = 201

Lag	LL	LR	df	p	FPE	AIC	HQIC	SBIC
0	-117.065				.189551	1.17478	1.18143	1.19121
1	620.726	1475.6*	1	0.000	.000124*	-6.15647*	-6.14317*	-6.12361*
2	621.255	1.0582	1	0.304	.000125	-6.15179	-6.13184	-6.10249
3	621.36	.20973	1	0.647	.000126	-6.14288	-6.11628	-6.07714
4	622.559	2.3994	1	0.121	.000126	-6.14487	-6.11162	-6.0627
5	622.845	.57051	1	0.450	.000126	-6.13776	-6.09786	-6.03915
6	622.848	.00639	1	0.936	.000128	-6.12784	-6.08129	-6.0128

* optimal lag
Endogenous: lc
Exogenous: _cons

Based on the Akaike Information Criterion (AIC), a lag of 1 appears to be the optimal choice for forecasting in this model.

```
. do "C:\Users\Windows\AppData\Local\Temp\STD29b8_000000.tmp"

. dfgls lc if $insample, maxlag(5) notrend

DF-GLS test for unit root                    Number of obs = 195
Variable: lc
Lag selection: User specified                Maximum lag   =   5

                                  ——— Critical value ———
 [lags]       DF-GLS mu         1%          5%          10%
─────────────────────────────────────────────────────────────
     5            2.711     -2.587      -2.013       -1.700
     4            3.097     -2.587      -2.018       -1.705
     3            3.304     -2.587      -2.023       -1.710
     2            4.215     -2.587      -2.028       -1.714
     1            5.330     -2.587      -2.032       -1.718
─────────────────────────────────────────────────────────────

Opt lag (Ng-Perron seq t) = 5 with RMSE = .0113941
Min SIC  = -8.819316 at lag 3 with RMSE = .0115192
Min MAIC = -8.772085 at lag 5 with RMSE = .0113941

.
end of do-file
```

The GF_GLS test statistic at lag 5 is 2.711. Based on this value and assuming conventional significance levels (1%), we can reject the null hypothesis (H_0) of a unit root being present in the data.

```
. dfuller lc if $insample, lags(0) regress

Dickey-Fuller test for unit root          Number of obs  = 201
Variable: lc                              Number of lags =   0

H0: Random walk without drift, d = 0

                                       Dickey-Fuller
                   Test      -------- critical value ---------
              statistic           1%           5%          10%
--------------------------------------------------------------
 Z(t)            -1.407       -3.476       -2.883       -2.573
--------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.5788.

Regression table
------------------------------------------------------------------------------
        D.lc | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
          lc |
         L1. |  -.0025343   .0018009    -1.41   0.161    -.0060856     .001017
             |
       _cons |   .0299688   .0160795     1.86   0.064    -.0017393     .061677
------------------------------------------------------------------------------

.
end of do-file
```

The Dickey-Fuller test statistic is -1.407. At a significance level (1%, 5%, or 10%), we failed to reject the null hypothesis (H_0) of a unit root presence in the data. (p_value = 0.5788)

- Augmented Dickey-Fuller (ADF) test failed to reject the null hypothesis (H_0) of a unit root presence at all significance levels. This suggests the data series in the model might still exhibit a unit root.

<u>VEC_part:</u>

The Augmented Dickey-Fuller (ADF) tests conducted on all variables (lc, ly, pi, r) failed to reject the null hypothesis of a unit root presence. This implies that the individual series likely exhibit non-stationary behavior (all of them are I(1)). Consequently, a Vector Error Correction Model (VECM) might be a suitable choice for forecasting, as it can accommodate non-stationary integrated variables.

Prior to estimating the Vector Error Correction Model (VECM), we can employ the vecrank function or code to determine the optimal cointegrating rank:

```
. vecrank r pi ly lc if $insample, trend(rconstant) lags(4) levela

Johansen tests for cointegration
Trend: Restricted constant                              Number of obs  = 201
Sample: 1971q3 thru 2021q3                              Number of lags =   4
---------------------------------------------------------------------------
Maximum                                     Trace     ---Critical value---
   rank  Params        LL  Eigenvalue   statistic          5%          1%
      0      48 1030.1943                 83.1981       53.12       60.16
      1      56 1049.6118     0.17569     44.3631       34.91       41.07
      2      62 1062.6788     0.12192     18.2290*1*5   19.96       24.60
      3      66 1068.9916     0.06088      5.6034        9.42       12.97
      4      68 1071.7933     0.02749
---------------------------------------------------------------------------
* selected rank
```

- Based on the information in this table, we fail to reject the null hypothesis (H_0) of a cointegrating rank of 2 at a 5% significant level. This suggests the presence of two long-run equilibrium relationships among the variables.

- Therefore, a Vector Error Correction Model appears suitable for capturing these relationships.

Before proceeding to the cointegration regression table and autocorrection analysis, it's recommended to determine the optimal lag length for the VAR model.

This can be achieved using tools like the Var_soc code which might employ information criteria like AIC to choose the optimal lag length.

```
. varsoc ly lc r pi if $insample, maxlag(4)

Lag-order selection criteria

   Sample: 1971q3 thru 2021q3                              Number of obs = 201
```

Lag	LL	LR	df	p	FPE	AIC	HQIC	SBIC
0	-291.912				.000223	2.9444	2.971	3.01014
1	969.852	2523.5	16	0.000	9.2e-10	-9.45126	-9.31826	-9.12258
2	1045.29	150.88	16	0.000	5.1e-10	-10.0427	-9.80328*	-9.45105*
3	1064.27	37.955*	16	0.002	5.0e-10*	-10.0723*	-9.72651	-9.21772
4	1071.79	15.053	16	0.521	5.4e-10	-9.98799	-9.53579	-8.87046

```
   * optimal lag
   Endogenous: ly lc r pi
    Exogenous: _cons

.
end of do-file

.
```

- Using AIC information criteria, we can choose lag 3 for the model VEC

<u>Cointegration Regression:</u>

We run the Cointegration Regression to produce **the regression table:**

```
. vec ly lc pi r if $insample, trend(rconstant) lags(3) rank(2)

Vector error-correction model

Sample: 1971q3 thru 2021q3                      Number of obs     =       201
                                                AIC               = -10.04998
Log likelihood =  1056.023                      HQIC              = -9.744076
Det(Sigma_ml)  =  3.21e-10                      SBIC              = -9.293998

Equation           Parms      RMSE     R-sq      chi2     P>chi2
----------------------------------------------------------------
D_ly                 10     .010429   0.3867   119.8101   0.0000
D_lc                 10     .010797   0.3747   113.8338   0.0000
D_pi                 10     .328854   0.3902   121.5642   0.0000
D_r                  10     1.17425   0.1682   38.43181   0.0000
----------------------------------------------------------------

------------------------------------------------------------------------------
             | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
D_ly         |
        _ce1 |
         L1. |  -.1209645   .0651818    -1.86   0.063    -.2487185    .0067896
             |
        _ce2 |
         L1. |   .1030875    .059973     1.72   0.086    -.0144574    .2206324
             |
          ly |
         LD. |  -.0987983   .1434085    -0.69   0.491    -.3798739    .1822773
        L2D. |  -.0423348   .1292016    -0.33   0.743    -.2955652    .2108956
             |
          lc |
         LD. |   .0833996   .1382557     0.60   0.546    -.1875766    .3543758
        L2D. |   .0468457   .1307868     0.36   0.720    -.2094918    .3031832
             |
          pi |
         LD. |  -.0048782   .0023715    -2.06   0.040    -.0095262   -.0002301
        L2D. |   .0045061   .0025271     1.78   0.075     -.000447    .0094591
             |
           r |
         LD. |   .0015932   .0006384     2.50   0.013      .000342    .0028443
        L2D. |  -.0002179   .0007007    -0.31   0.756    -.0015912    .0011554
-------------+----------------------------------------------------------------
D_lc         |
        _ce1 |
         L1. |   .0131052   .0674832     0.19   0.846    -.1191595    .1453698
             |
        _ce2 |
         L1. |  -.0212379   .0620904    -0.34   0.732     -.142933    .1004571
             |
          ly |
         LD. |   .0034301   .1484718     0.02   0.982    -.2875694    .2944295
        L2D. |   .0061462   .1337632     0.05   0.963    -.2560249    .2683174
             |
          lc |
         LD. |  -.0764436   .1431371    -0.53   0.593    -.3569871    .2040998
        L2D. |    .001785   .1354045     0.01   0.989    -.2636029     .267173
             |
          pi |
         LD. |  -.0037573   .0024552    -1.53   0.126    -.0085695    .0010549
```

L2D.	.001785	.1354045	0.01	0.989	-.2636029	.267173
pi						
LD.	-.0037573	.0024552	-1.53	0.126	-.0085695	.0010549
L2D.	.002349	.0026163	0.90	0.369	-.0027789	.0074769
r						
LD.	-.0002356	.0006609	-0.36	0.722	-.0015309	.0010598
L2D.	-.0007372	.0007254	-1.02	0.310	-.002159	.0006846
D_pi						
_ce1						
L1.	-5.228233	2.055311	-2.54	0.011	-9.256568	-1.199897
_ce2						
L1.	4.749679	1.891066	2.51	0.012	1.043258	8.456101
ly						
LD.	5.48727	4.521953	1.21	0.225	-3.375595	14.35013
L2D.	3.24782	4.073979	0.80	0.425	-4.737031	11.23267
lc						
LD.	-7.920148	4.359474	-1.82	0.069	-16.46456	.624263
L2D.	-4.716941	4.123966	-1.14	0.253	-12.79977	3.365884
pi						
LD.	.4951192	.0747784	6.62	0.000	.3485562	.6416823
L2D.	.1537711	.0796849	1.93	0.054	-.0024084	.3099505
r						
LD.	.0608955	.0201286	3.03	0.002	.0214441	.1003468
L2D.	.0044406	.0220935	0.20	0.841	-.0388619	.0477432
D_r						

D_r						
_ce1						
L1.	-5.730854	7.338984	-0.78	0.435	-20.115	8.653289
_ce2						
L1.	5.44478	6.752508	0.81	0.420	-7.789892	18.67945
ly						
LD.	45.29721	16.14672	2.81	0.005	13.65022	76.94421
L2D.	47.69545	14.54712	3.28	0.001	19.18361	76.20728
lc						
LD.	-24.00719	15.56655	-1.54	0.123	-54.51707	6.502689
L2D.	-40.16983	14.72561	-2.73	0.006	-69.0315	-11.30816
pi						
LD.	.0897444	.2670145	0.34	0.737	-.4335943	.6130831
L2D.	.6098749	.284534	2.14	0.032	.0521984	1.167551
r						
LD.	-.2562546	.071874	-3.57	0.000	-.3971251	-.1153841
L2D.	-.3366787	.0788903	-4.27	0.000	-.4913009	-.1820565

Cointegrating equations

Equation	Parms	chi2	P>chi2
_ce1	2	28.59722	0.0000
_ce2	2	27.19929	0.0000

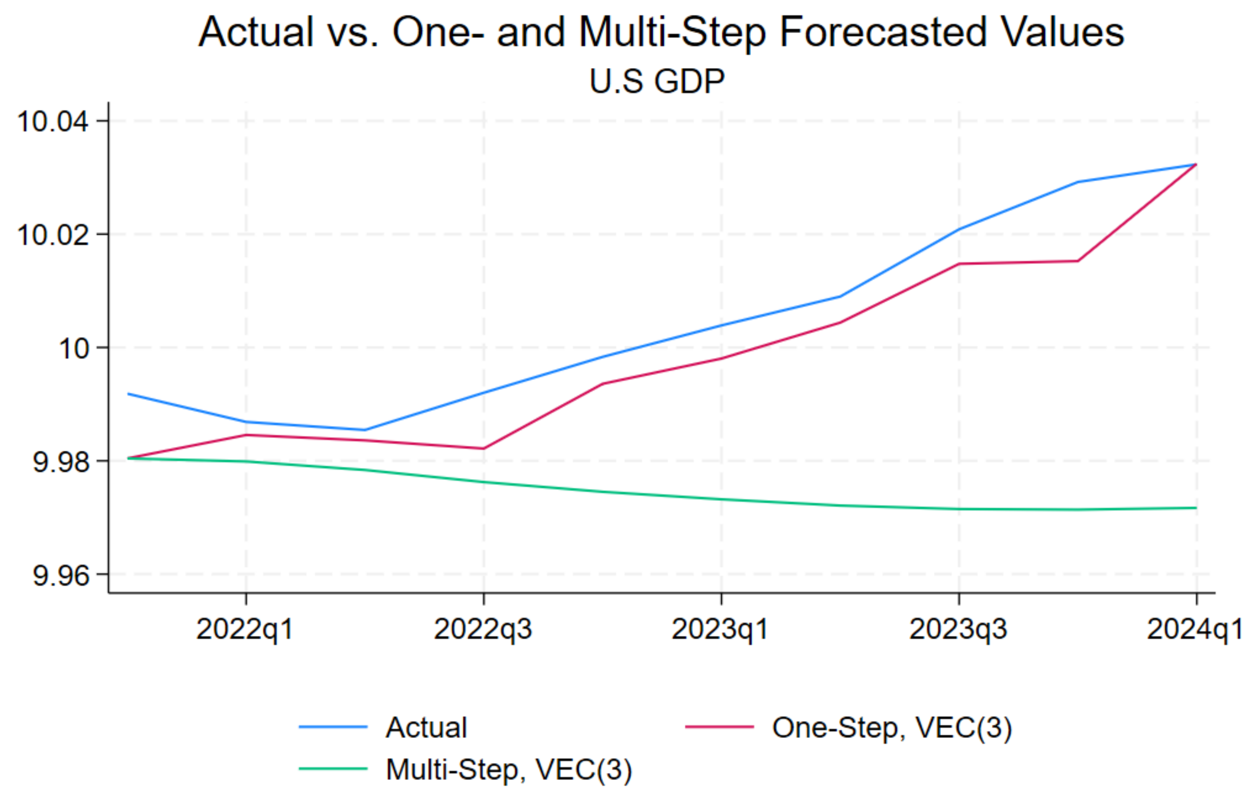
Identification: beta is exactly identified

Johansen normalization restrictions imposed

beta	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
_ce1						
ly	1	.	.	.	.	.
lc	-4.44e-16	.	.	.	.	.
pi	.1340949	.048924	2.74	0.006	.0382057	.2299842
r	.0260958	.0282557	0.92	0.356	-.0292844	.0814761
_cons	-10.71972	.1506448	-71.16	0.000	-11.01498	-10.42447
_ce2						
ly	0	(omitted)				
lc	1	.	.	.	.	.
pi	.1379173	.0532812	2.59	0.010	.0334881	.2423464
r	.0306804	.0307722	1.00	0.319	-.0296321	.0909928
_cons	-10.39245	.1640613	-63.34	0.000	-10.71401	-10.0709

<u>In Sample – Out Sample Forcast:</u>

***Long- Period Forcast:**

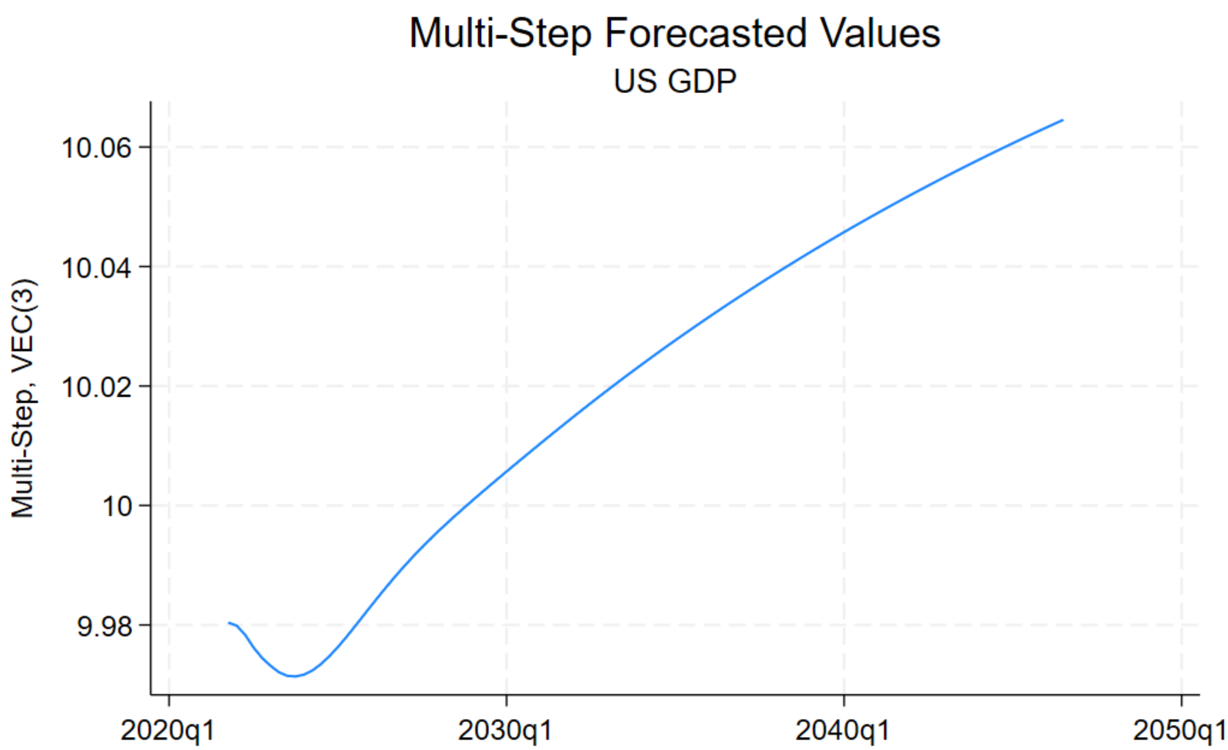


Comment:

While a VEC (3) – rank 2 model can be used to estimate both one-step and multi-step forecasts, it's important to consider that VECMs are not designed to capture ***mean reversion***. This means that multi-step forecasts generated by the VECM might not necessarily return to the mean over time.

Additionally, the forecasts, particularly multi-step ones, could be influenced by the last observation in the model, including inflation, real output GDP, real consumption expenditure, and interest rate. Three last observations of the variables will affect the forecasted outcome in the long run.

<u>Multi-Step Long-ahead for VEC (3) model:</u>



Comment:

The last observation of Output – GDP (~9.9750 – in 2021q3) of in sample period (potentially linked to decreased consumption or the effects of rising inflation) might influence the VECM's forecasts, especially for multiple-step long-run ahead.

Question 6:

Log data density (Modified Harmonic Mean) is -231.387588.

parameters

	prior mean	post. mean	90% HPD interval		prior	pstdev
theta	0.600	0.0500	0.0466	0.0531	norm	0.1500
sigma	2.500	2.6054	2.5894	2.6255	norm	0.2500
phi_pi	1.500	2.2843	2.2630	2.3091	norm	0.2500
phi_y	0.125	0.2184	0.2026	0.2305	norm	0.1000
rho_r	0.750	0.8131	0.7922	0.8370	beta	0.1000
rho_z	0.750	0.9761	0.9626	0.9911	beta	0.1000
rho_u	0.750	0.9299	0.9190	0.9374	beta	0.1000
rho_m	0.500	0.0504	0.0291	0.0693	beta	0.2000
rho_g	0.500	0.9405	0.9099	0.9614	beta	0.2000
sc	0.750	0.8476	0.8337	0.8605	norm	0.1000
sg	0.750	0.1145	0.1139	0.1152	norm	0.1000

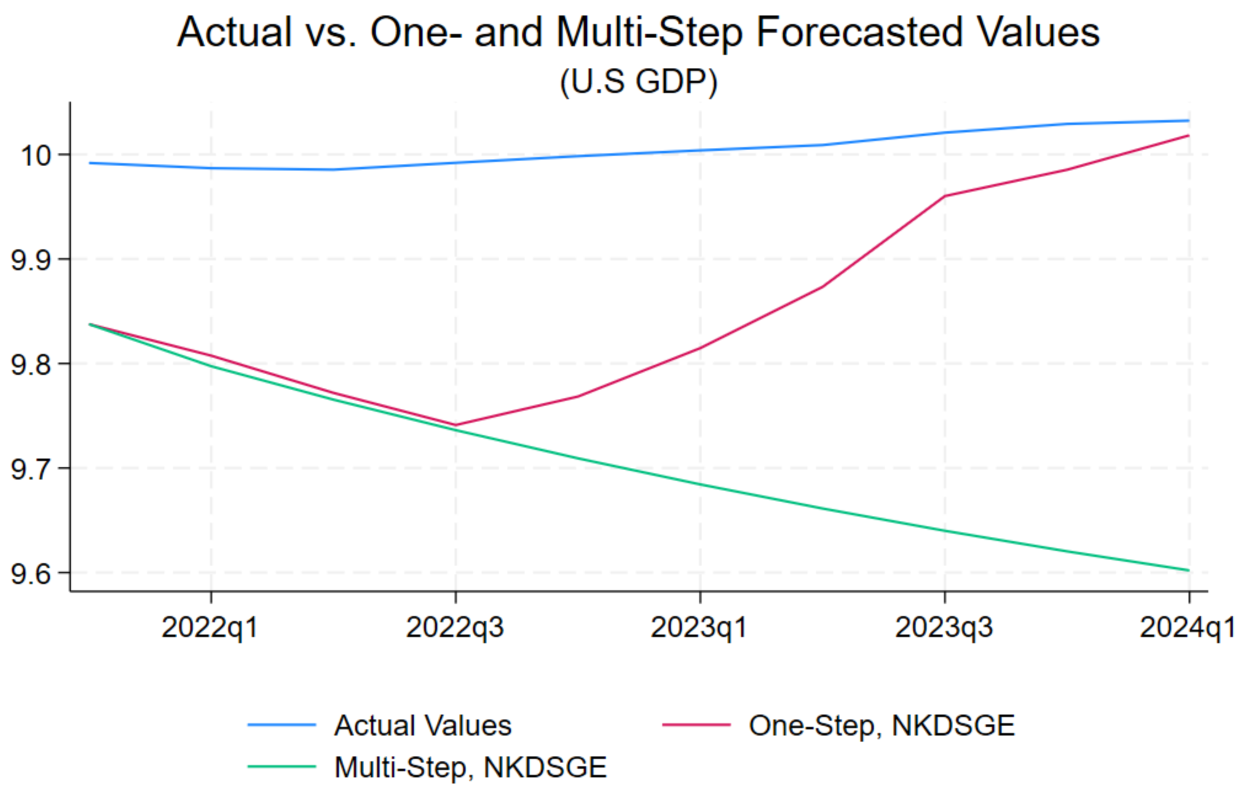
standard deviation of shocks

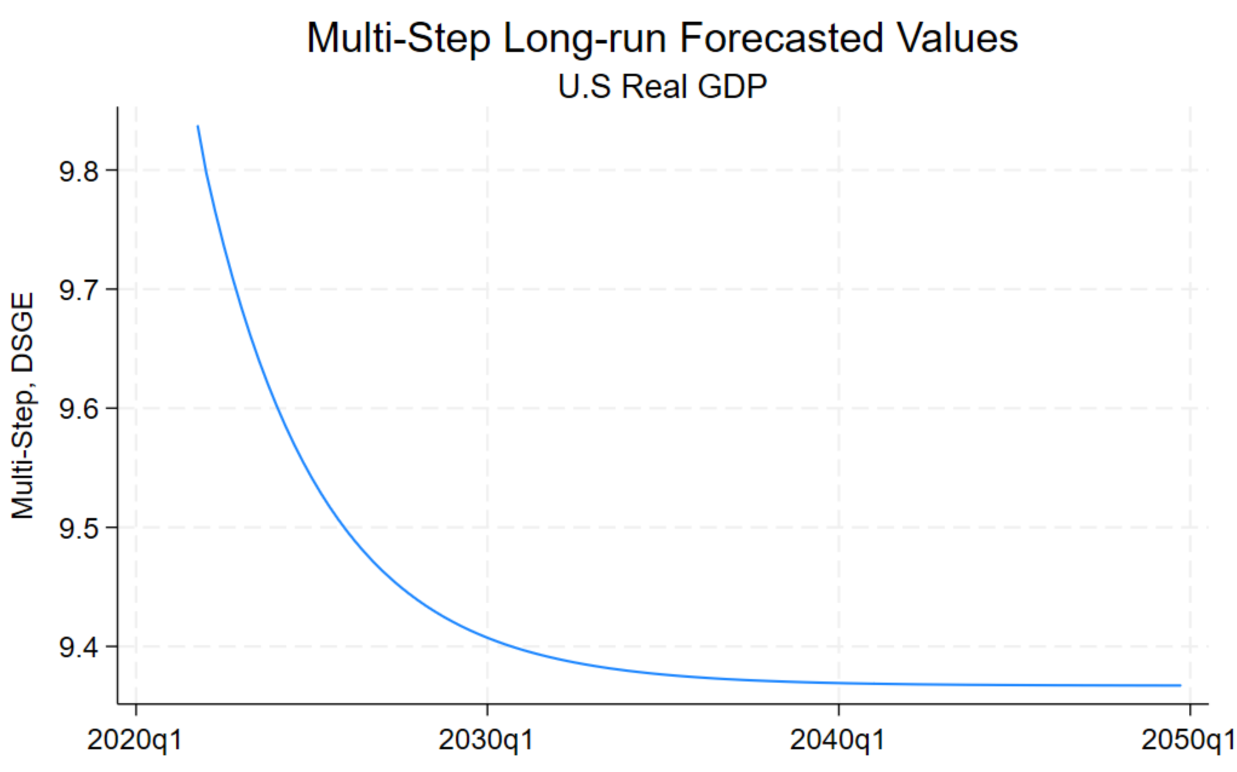
	prior mean	post. mean	90% HPD interval		prior	pstdev
e_z	0.050	1.3223	1.2568	1.3837	invg	2.0000
e_u	0.050	0.4127	0.4015	0.4266	invg	2.0000
e_m	0.050	1.1793	1.0763	1.2457	invg	2.0000
e_g	0.875	0.2840	0.2657	0.2978	invg	0.4300

Some Explanations: I employed Bayesian Maximum Likelihood (BML) estimation for a DSGE model in Dynare. To establish the prior mean for the model parameters, I reviewed relevant economic journals (references provided) to determine appropriate prior means and standard deviations. For example, I chose rho_g with Beta distribution (mean - 0.500, standard deviation-0.200) – according to Paccagnini, et al. (2014), and some declared values in Dynare form Costantini, et al.(2010)

Additionally, I also incorporated prior mean information from the course lectures and problemsets for specific coefficients (with each variable in the table).

Some tables/graphs for the distribution of the shocks (Dynare estimation) were attached to the Appendix

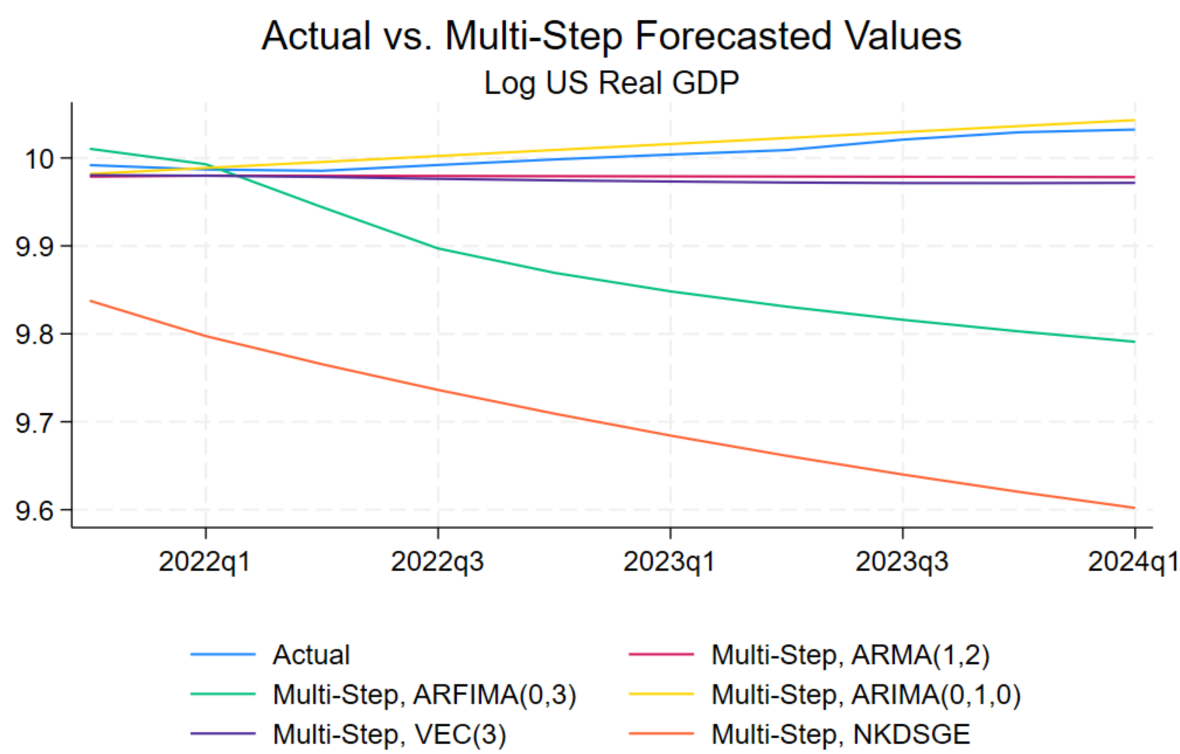

Actual vs. One- and Multi-Step Forecasted Values
(U.S GDP)
10
9.9
9.8
9.7
9.6
2022q1
2022q3
2023q1
2023q3
2024q1
Actual Values
One-Step, NKDSGE
Multi-Step, NKDSGE

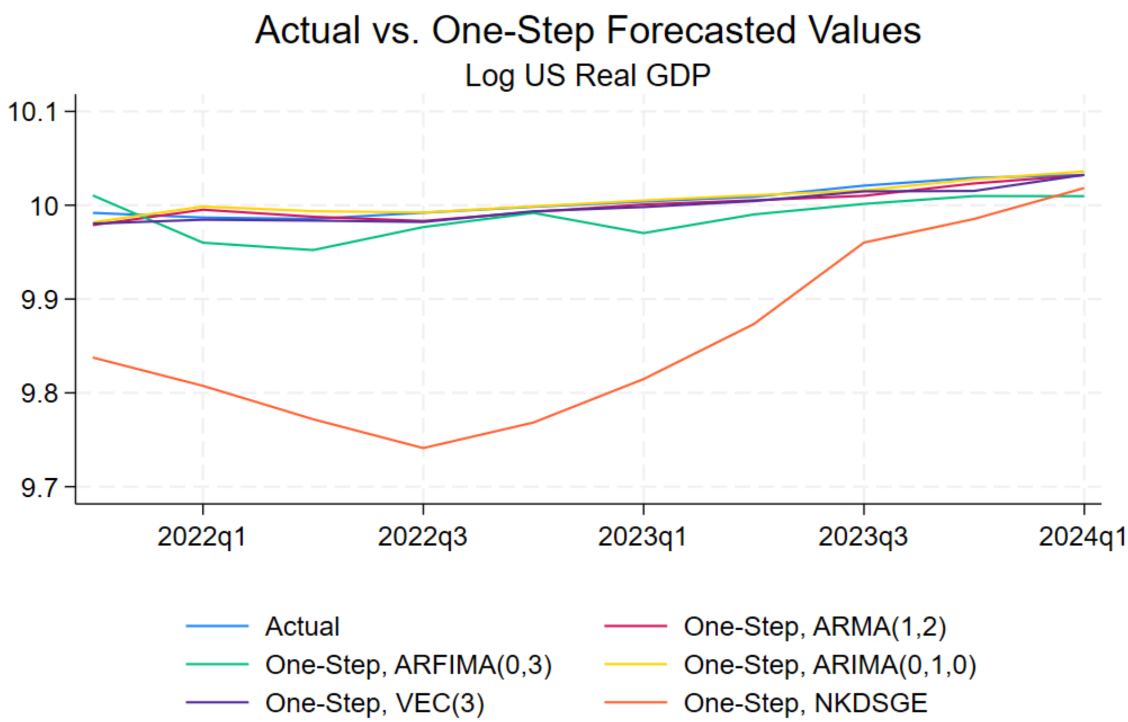

Multi-Step Long-run Forecasted Values
U.S Real GDP
Multi-Step, DSGE
9.8
9.7
9.6
9.5
9.4
2020q1
2030q1
2040q1
2050q1

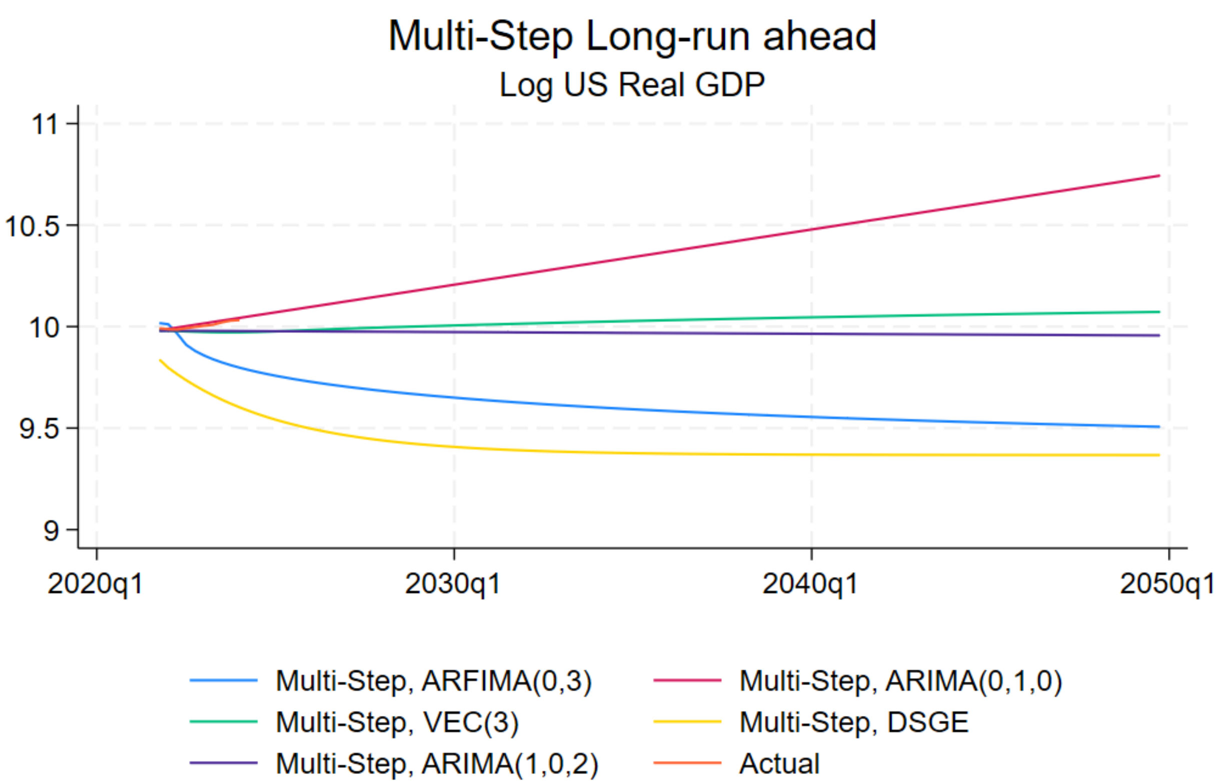
Comment on features of the DSGE forecast:

- The forecast accuracy has improved following adjustments made to the coefficient distributions of certain variables.
- Concerningly, the one-step forecast consistently falls short of the actual data(out-sample) data from 2022q1 until they can be converged in 2024q2.
- The multi-step forecast exhibits a downward trend. While this trend might potentially revert to the mean of sample data as we extend the forecast horizon to 2050, it's important to remember that DSGE models are generally less reliable for long-term predictions.
- The in-sample data suggests that Output - GDP might follow a random walk with drift. This characteristic can pose challenges for DSGE models, as they may not perfectly capture all the short-term fluctuations in the data.

Question 7:




Actual vs. One-Step Forecasted Values
Log US Real GDP
10.1
10
9.9
9.8
9.7
2022q1
2022q3
2023q1
2023q3
2024q1
Actual
One-Step, ARFIMA(0,3)
One-Step, VEC(3)
One-Step, ARMA(1,2)
One-Step, ARIMA(0,1,0)
One-Step, NKDSGE


Multi-Step Long-run ahead
Log US Real GDP
11
10.5
10
9.5
9
2020q1
2030q1
2040q1
2050q1
Multi-Step, ARFIMA(0,3)
Multi-Step, VEC(3)
Multi-Step, ARIMA(1,0,2)
Multi-Step, ARIMA(0,1,0)
Multi-Step, DSGE
Actual

```
. summarize rmsearma rmsearfima rmsearima rmsevec rmsedsge
```

Variable	Obs	Mean	Std. dev.	Min	Max
rmsearma	10	.0061865	.0040437	.0002252	.0130504
rmsearfima	10	.0214345	.0081726	.006511	.0335991
rmsearima	10	.0044006	.0042141	.000238	.0117918
rmsevec	10	.0060785	.0044324	.0001154	.0139647
rmsedsge	10	.1471254	.0822495	.0139942	.2508621

In the case of One-Step Forecasting:

- Based on the root mean square error (RMSE) comparisons, the ARIMA(0,1,0) model demonstrates the best performance for one-step forecasting in this case.
- **DSGE Model Limitations:** The DSGE model exhibits the highest RMSE, indicating its limitations in capturing short-term dynamics for one-step forecasts within this specific scenario.

In the case of Multi-Step Forcasting:

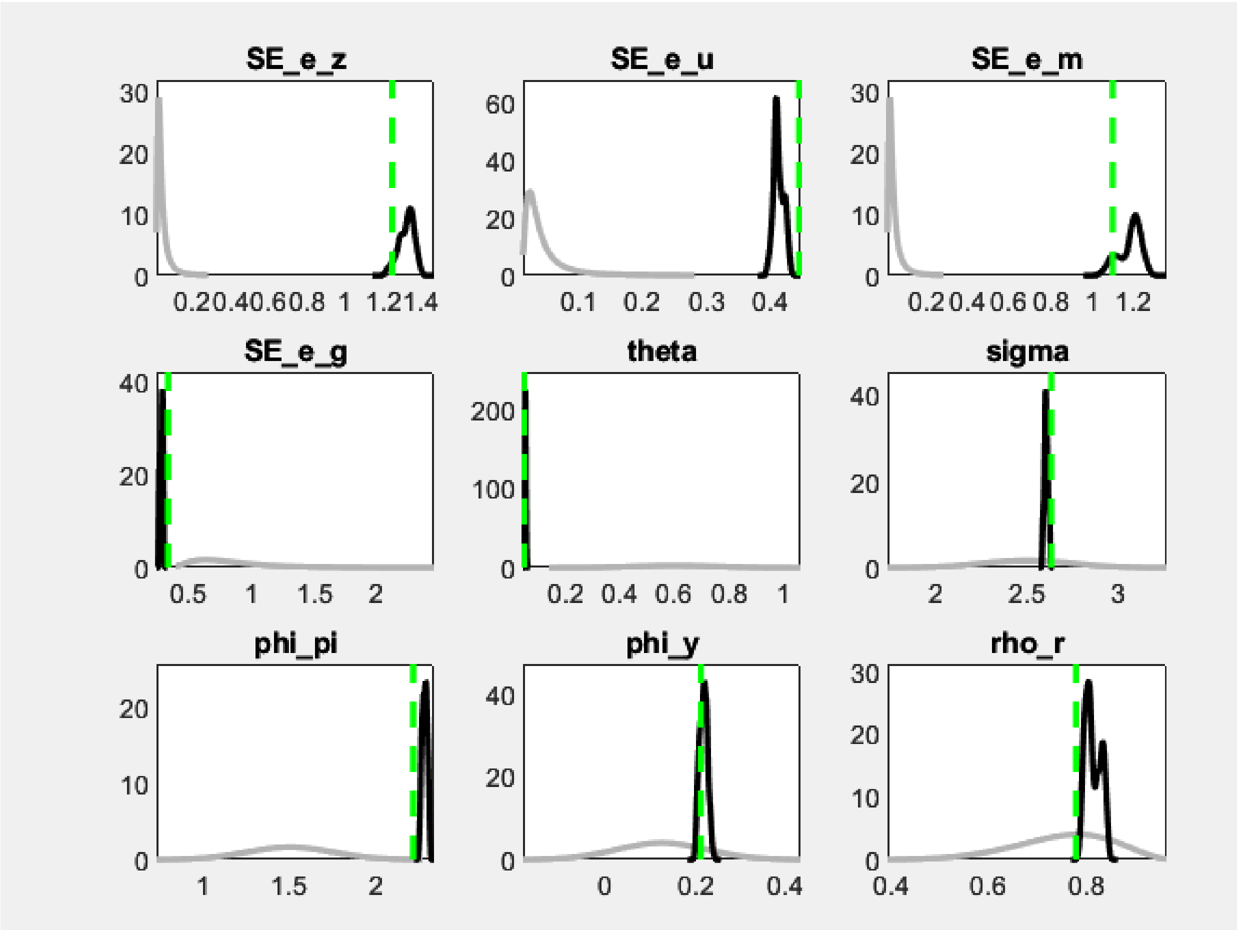
- We can also conclude that the model **ARIMA(0,1,0)** is still the best model to forecast in the form of multi-step. This suggests that the model of random walk with positive drift is a good model for forecasting real GDP in this case.

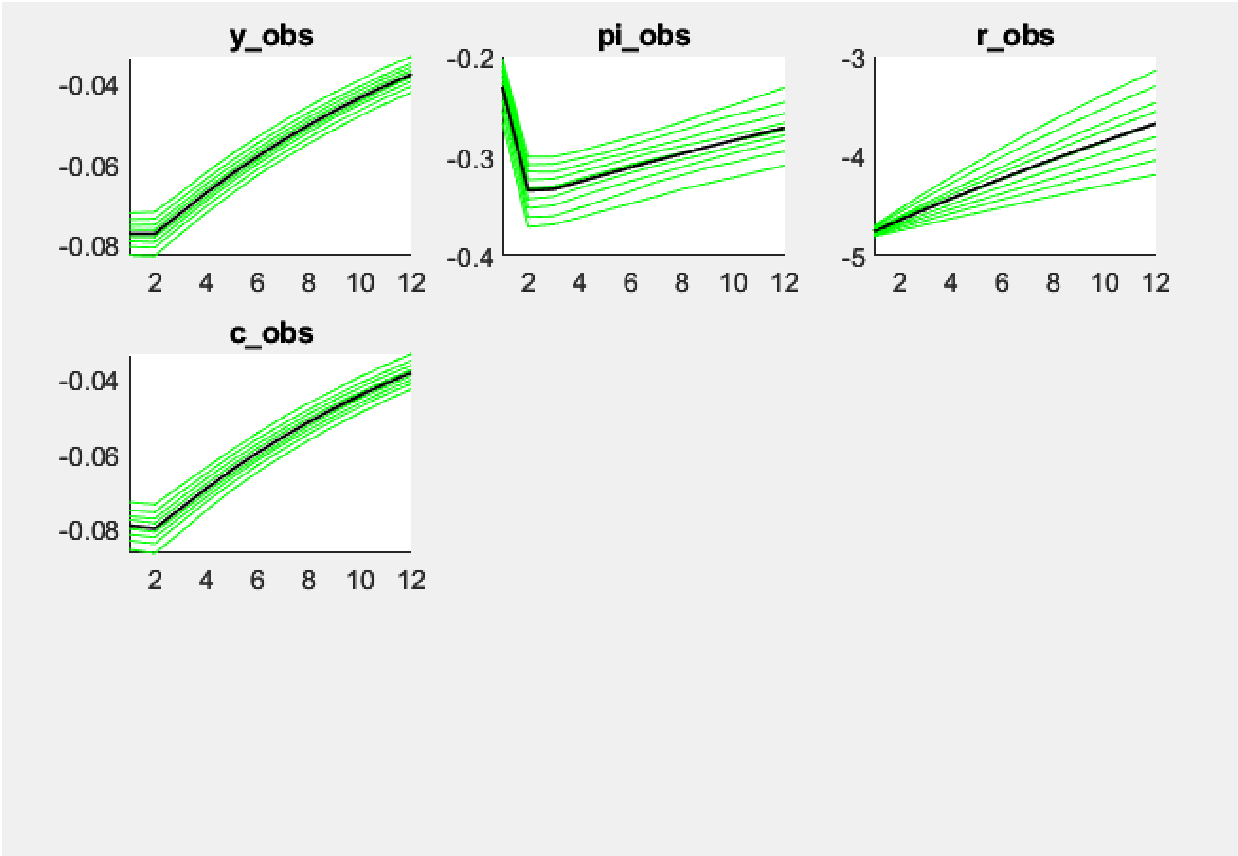
REFERENCE:

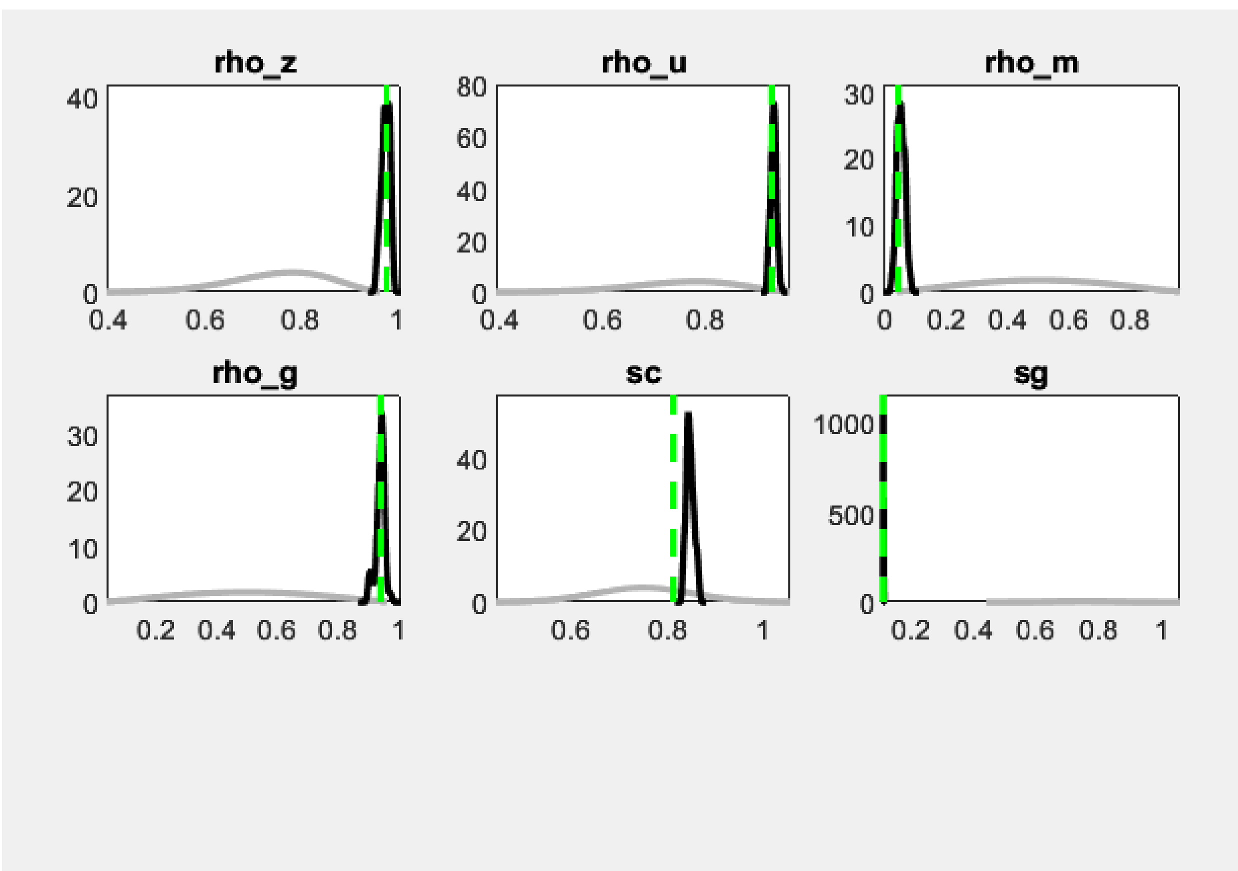
Costantini, Mauro & Gunter, Ulrich & Kunst, Robert. (2010). Forecast Combination Based on Multiple Encompassing Tests in a Macroeconomic DSGE System. Institute for Advanced Studies, Economics Series.

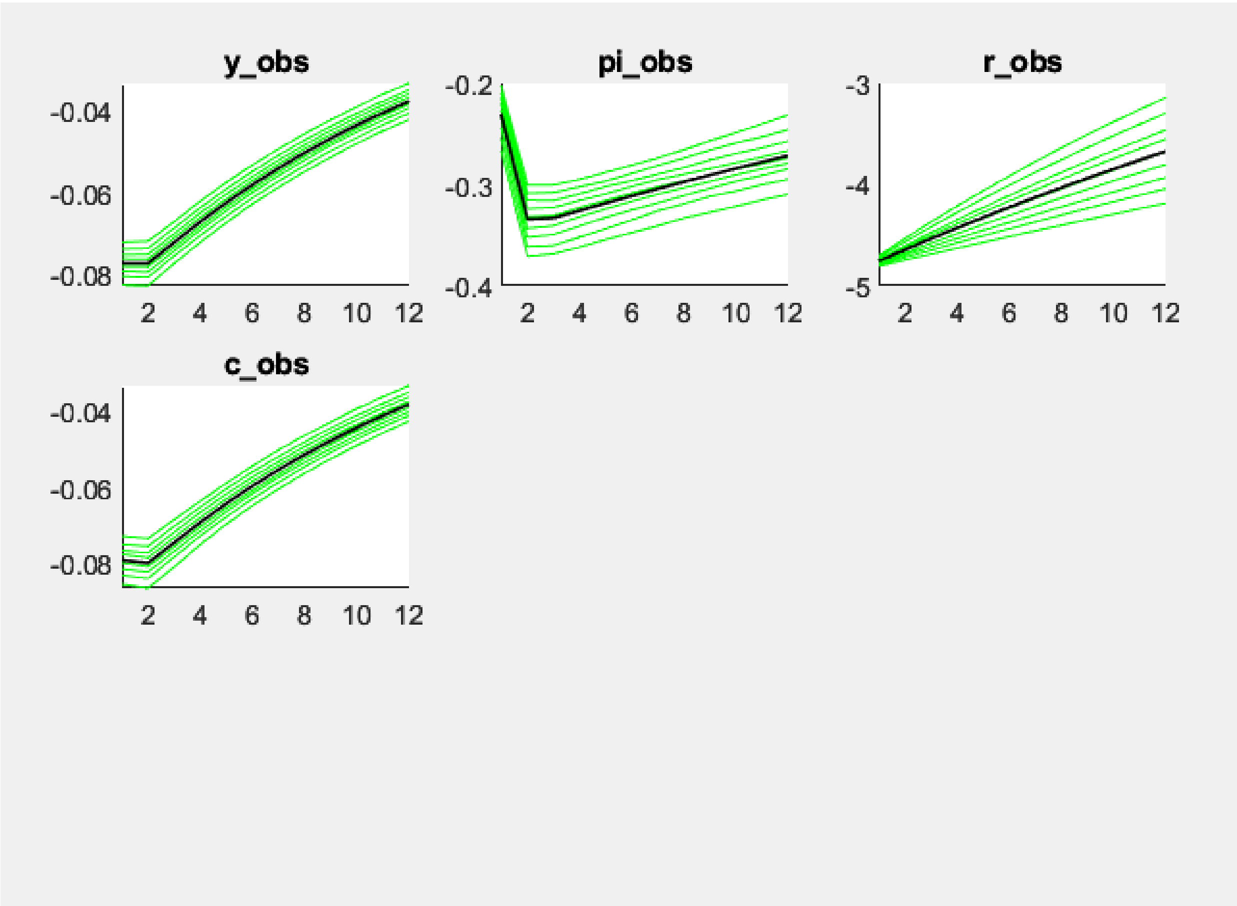
Paccagnini, Alessia & Bekiros, Stelios. (2014). Macroprudential policy and forecasting using hybrid dsge models with financial frictions and state space markov-switching TVP-VARs. Macroeconomic Dynamics. 19. 10.1017/S1365100513000953.

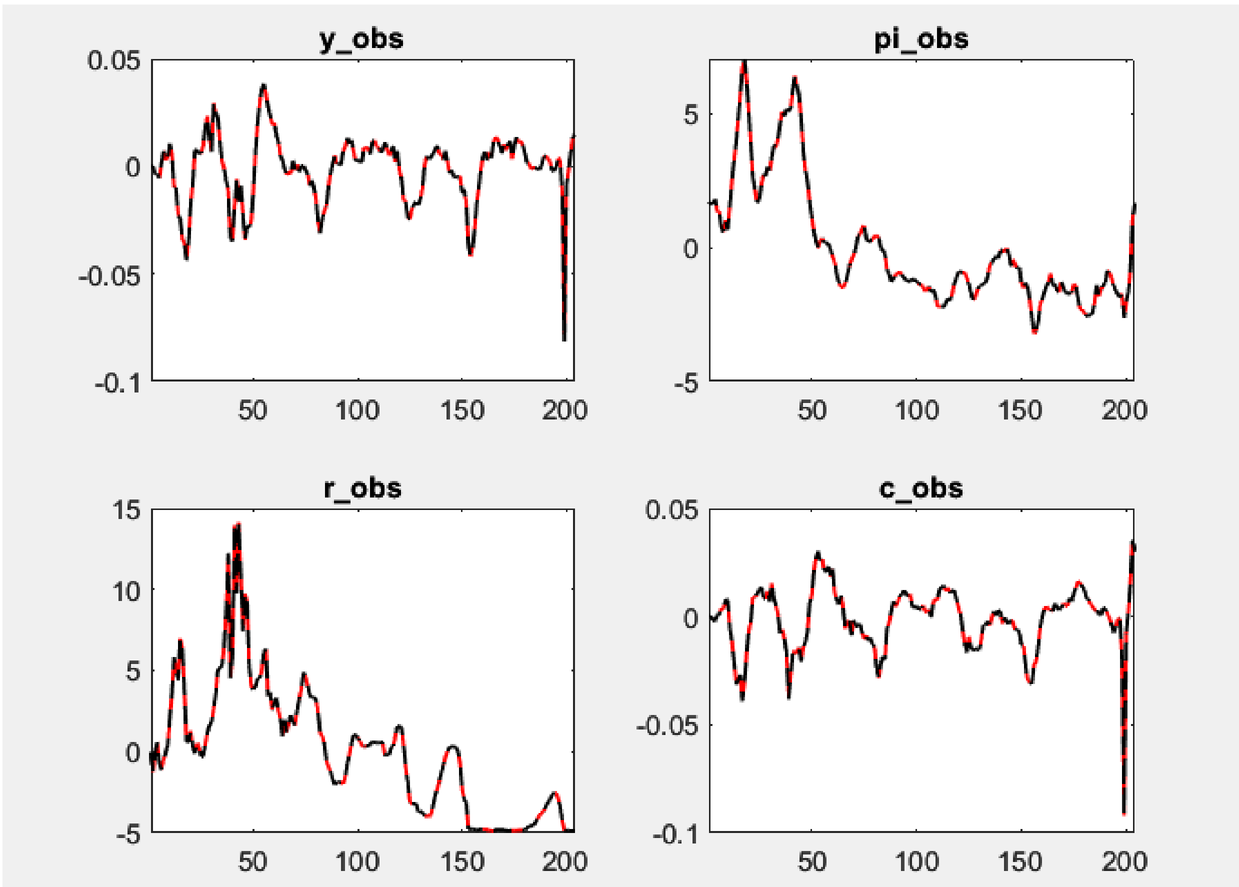
APPENDIX: DYNARE/MATLAB ESTIMATION GRAPH



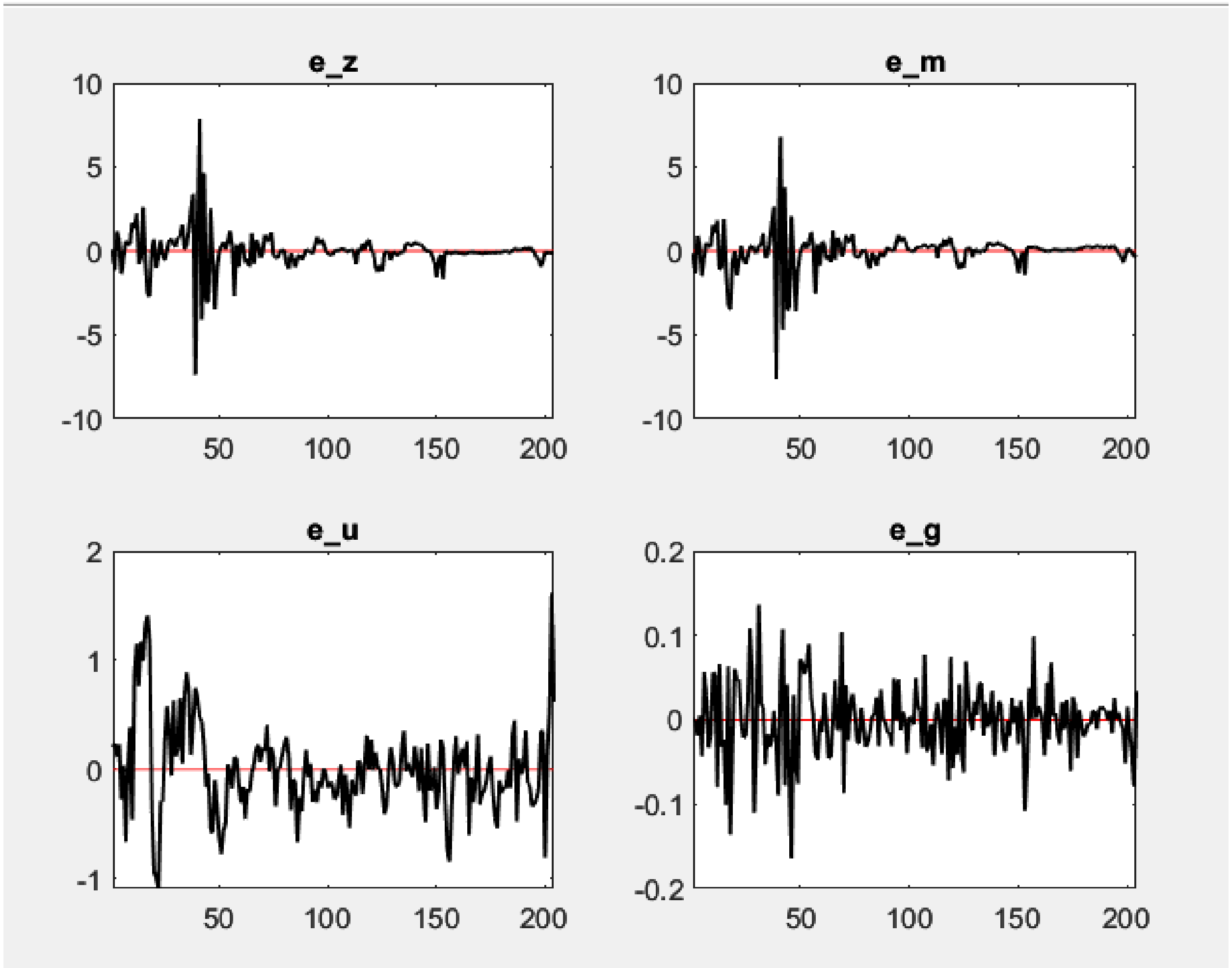

y_obs
pi_obs
r_obs
c_obs
-0.04
-0.06
-0.08
-0.2
-0.3
-0.4
-3
-4
-5
2 4 6 8 10 12


rho_z
rho_u
rho_m
rho_g
sc
sg
40
20
0
80
60
30
10
1000
500
0.4 0.6 0.8 1
0 0.2 0.4 0.6 0.8
0.2 0.4 0.6 0.8 1
0.6 0.8 1


y_obs
pi_obs
r_obs
c_obs
-0.04
-0.06
-0.08
-0.2
-0.3
-0.4
-3
-4
-5
2
4
6
8
10
12



y_obs
0.05
0
-0.05
-0.1
50
100
150
200
pi_obs
5
0
-5
50
100
150
200
r_obs
15
10
5
0
-5
50
100
150
200
c_obs
0.05
0
-0.05
-0.1
50
100
150
200


e_z
e_m
e_u
e_g

RESULTS FROM POSTERIOR ESTIMATION

parameters

	prior mean	mode	s.d.	prior	pstdev
theta	0.6000	0.0440	0.0498	norm	0.1500
sigma	2.5000	2.6339	0.0444	norm	0.2500
phi_pi	1.5000	2.2177	0.0392	norm	0.2500
phi_y	0.1250	0.2123	0.0171	norm	0.1000
rho_r	0.7500	0.7788	0.0097	beta	0.1000
rho_z	0.7500	0.9779	0.0218	beta	0.1000
rho_u	0.7500	0.9267	0.0174	beta	0.1000
rho_m	0.5000	0.0421	0.0216	beta	0.2000
rho_g	0.5000	0.9407	0.0184	beta	0.2000
sc	0.7500	0.8131	0.0084	norm	0.1000
sg	0.7500	0.1139	0.0045	norm	0.1000

standard deviation of shocks

	prior mean	mode	s.d.	prior	pstdev
e_z	0.0500	1.2481	0.0574	invg	2.0000
e_u	0.0500	0.4446	0.0181	invg	2.0000
e_m	0.0500	1.0971	0.0537	invg	2.0000
e_g	0.8750	0.3377	0.0332	invg	0.4300

4

MCMC Inefficiency factors per block

Parameter	Block 1
SE_e_z	497.205
SE_e_u	502.716
SE_e_m	590.444
SE_e_g	633.928
theta	146.697
sigma	567.141
phi_pi	512.777
phi_y	616.572
rho_r	662.480
rho_z	489.757
rho_u	471.683
rho_m	530.937
rho_g	522.256
sc	572.452
sg	67.908

Geweke (1992) Convergence Tests, based on means of draws 10000 to 12000 vs 15000 to 20000 for chain 1.
p-values are for Chi2-test for equality of means.

Parameter	Post. Mean	Post. Std	p-val No Taper	p-val 4% Taper	p-val 8% Taper	p-val 15% Taper
SE_e_z	1.318	0.044	0.000	0.005	0.034	0.087
SE_e_u	0.416	0.018	0.000	0.418	0.555	0.646
SE_e_m	1.151	0.064	0.000	0.000	0.001	0.008
SE_e_g	0.302	0.022	0.000	0.000	0.000	0.000
theta	0.049	0.002	0.000	0.002	0.011	0.034
sigma	2.622	0.019	0.000	0.000	0.000	0.002
phi_pi	2.260	0.030	0.000	0.000	0.000	0.000
phi_y	0.217	0.007	0.000	0.000	0.000	0.000
rho_r	0.802	0.017	0.000	0.000	0.000	0.000
rho_z	0.976	0.008	0.000	0.014	0.063	0.132
rho_u	0.927	0.006	0.000	0.033	0.116	0.215
rho_m	0.049	0.012	0.000	0.024	0.090	0.168
rho_g	0.939	0.020	0.000	0.000	0.001	0.010
sc	0.838	0.012	0.000	0.000	0.000	0.000
sg	0.115	0.001	0.000	0.416	0.445	0.458